Introduction of OpenF2A

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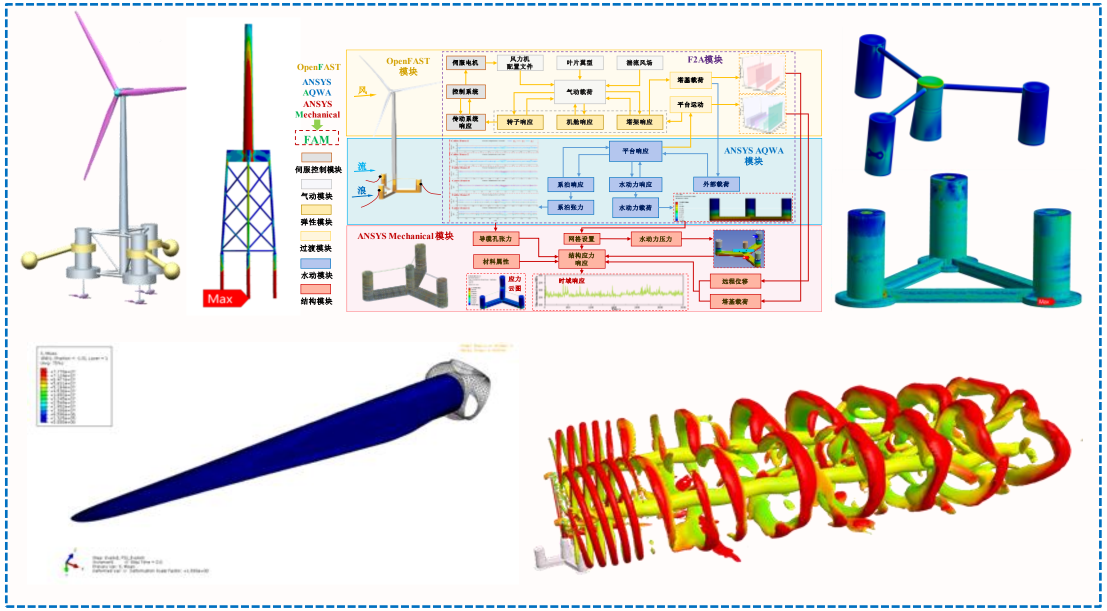
Contents

1. Backgroud

CV of Yang Yang

Yang Yang

- 2009.09-2020.04, *BsC*, *PhD*, ***University of Shanghai for Science and Technology***, P.R.China
- 2017.10-2018.10, Visiting *PhD* funded by *CSC*, ***Liverpool John Moores University***, UK
- 2018.10-2020.11, Post-doctoral Research Associate, ***Liverpool John Moores University***, UK
- 2021.05-2024.12, Associate Professor, ***Ningbo University***, P.R. China
- 2024.12-2025.09, Professor, ***Ningbo University***, P.R. China

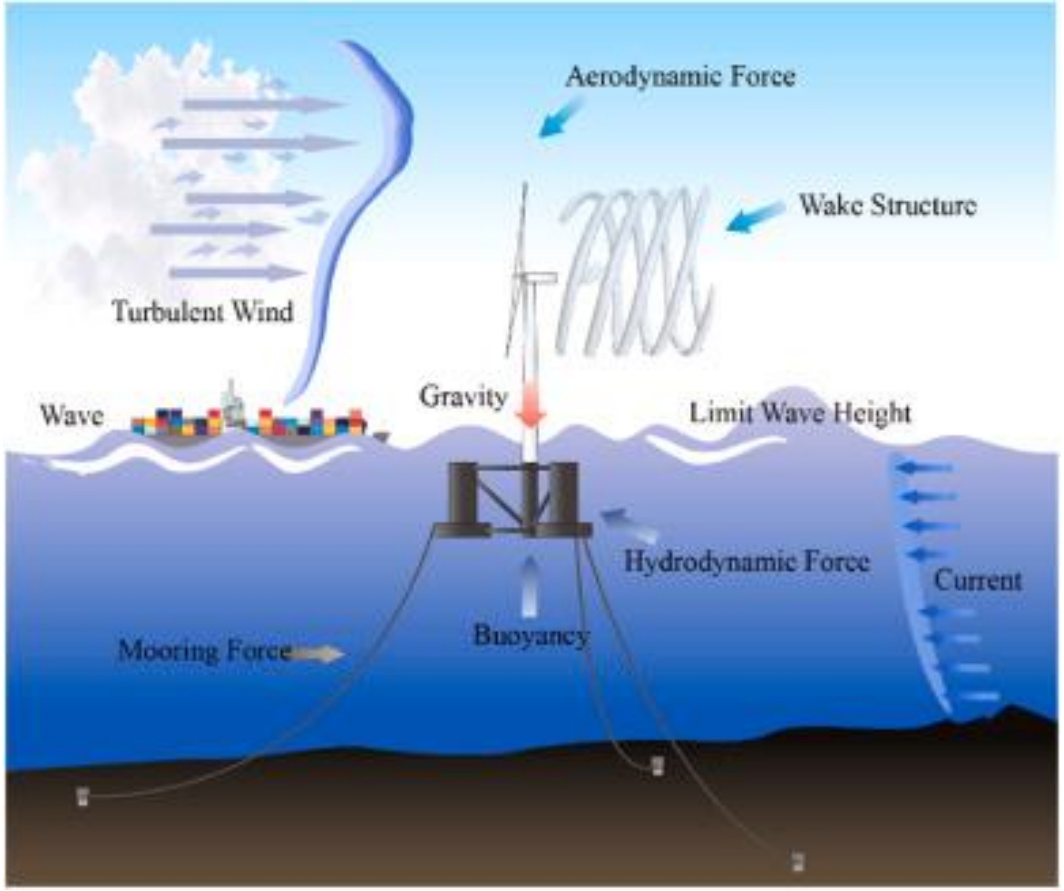


Research Interests

- Numerical modelling of wind-wave-current integrated floating energy systems
- Numerical modelling of twin-rotor floating offshore wind turbines
- Structural design of floating offshore wind platforms
- Aeroelastic modelling of offshore wind turbine blades
- Seismic analysis and structural control

https://scholar.google.com/citations?user=frTAwNQAAAAJ&hl=zh-CN

Fully coupled effects of FOWTs



Aerodynamic load
(Blades & Tower)

Elastic deformation
(Blades & Tower)

Dynamic responses

Servo-control
(Rotor speed & blade pitch)

Hydrodynamic loads and mooring restoring loads
(Platform)

Aero-Hydro-Servo-Elastic Coupling of Floating OWTs

1. Backgroud

Main numerical tools

Tool	Developer	Notes
Bladed	DNV in Norway	Earliest commercial software for wind turbine dynamic analysis,
SESAM	DNV/Sintef in Norway	Almost the only option for floating wind platform structural design
HAWC2	DTU in Denmark	One of the earliest tool, FEM for structures, without GUI, reliable aero-elastic
OrcaFlex	Orcina in UK	Reliable simulation tool for cables, and mooring lines, getting stronger
DeepLines Wind	PRINCIPIA in France	Suitable for VAWT/HAWT, FEM for structural analysis
Qblade	BTU in Germany	Open-source, FVW involved for aerodynamics, powerful for different FOWTs
Simpack	SIMULIA in France	Coupled with many OpenFAST modules, capable of modelling fractured blades
Ashes	Simis in Norway	Developed by graduates from NTNU, with good GUI
3DFloat	IFE in Norway	FEM for structural analysis, excellent in pile-soil interaction
OpenFAST	NREL in US	Most popular open-source numerical tool, widely used in academic area, still being developed
Samcef	Samtech in Belgium	Developed by SAMTECH, capable of modeling nonlinear pile-soil interaction

Main numerical tools

MI&T TimeFloat-FAST

Developed by MI&T by integrating FAST into hydrodynamic analysis tool TimeFloat. Dynamic response of the wind turbine are solved in FAST, responses of the mooring system and floating platform are predicted in TimeFloat.

Used for the design of WindFloat concept

Texas A&M CHARM3D-FAST

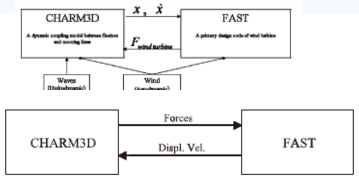


Fig. 3. Basic concept of CHARM3D-FAST hybrid model.

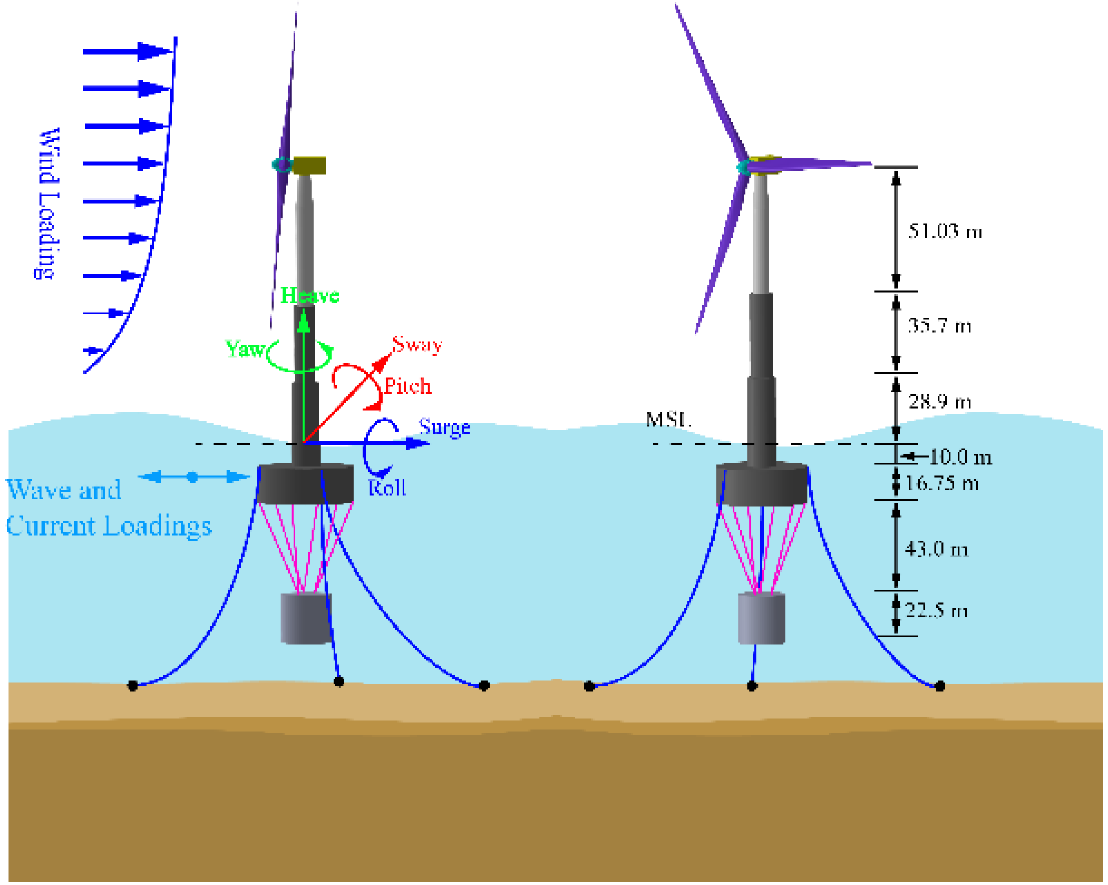
Similar to TimeFloat-FAST, this tool consists of a hydrodynamic analysis tool and FAST. It was originally developed by a master student in Texas A&M University in 2007 based on FAST v6. Displacement and velocity from CHARM3D are transferred to FAST and aerodynamics from FAST are fed back. Acceleration of the platform is ignored. However, the coupling relationship between FAST and CHARM3D in its later application is in the contrast to that was implemented previously.

Yang YANG FAST2AQWA(F2A)

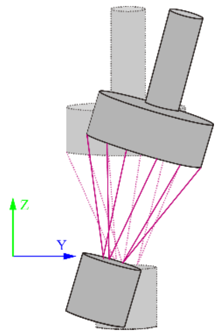
F2A is developed by Dr Yang Yang in Liverpool John Moores University for fully coupled analysis of a multi-body floating wind platform. Now it is released to the public on GitHub: https://github.com/yang7857854/F2A.

Similar to TimeFloat, dynamic responses of the wind turbine are predicted in OpenFAST and the responses of the platform and mooring lines are solved in AQWA.

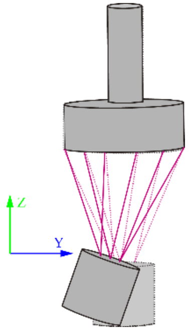
Motivation of development of F2A



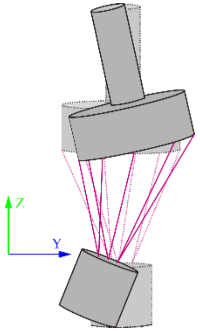
Telwind concept



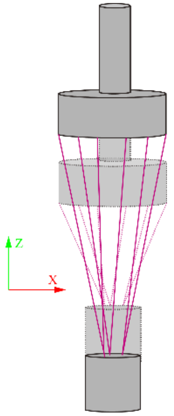
Collective pitch
36.5 s



Local pitch
3.08 s



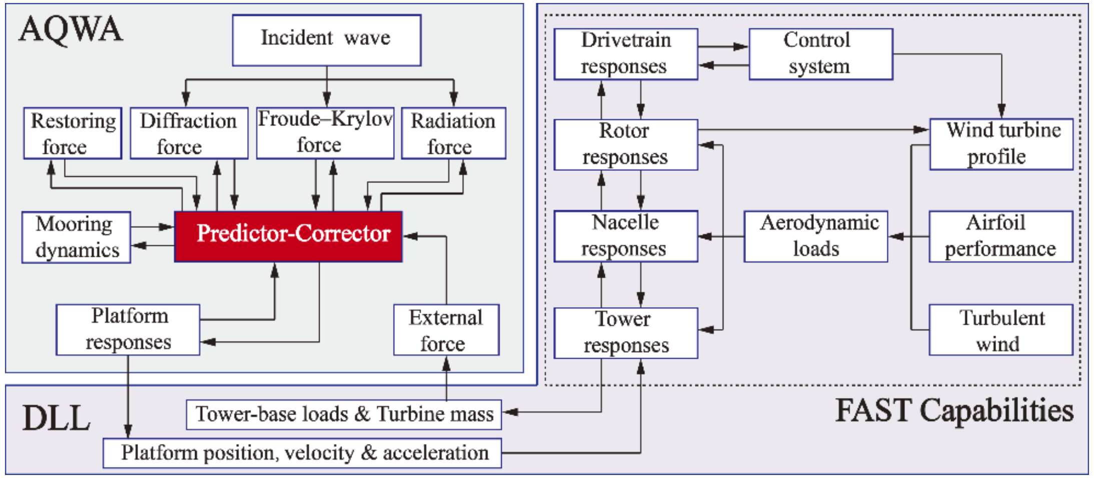
Coupling pitch
1.15 s



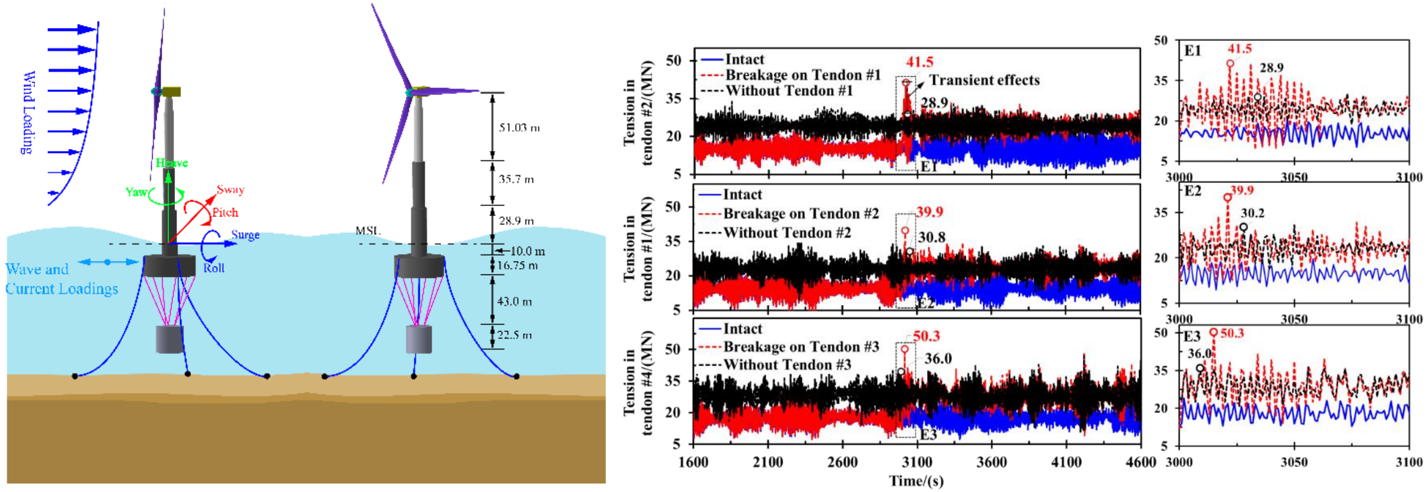
Relative heave
0.71 s

1. Backgroud

Development of F2A and its limitation

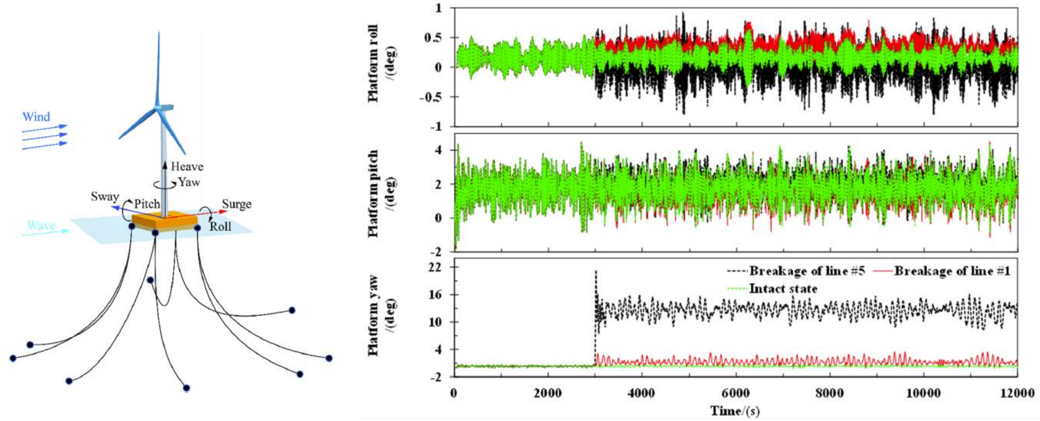


Framework of F2A (<u>Yang Y</u>, Bashir M, Michailides C, et al. Renewable Energy, 2020, 161: 606-625.)



Application for the Telwind concept

<u>Yang Y</u>, Bashir M, Michailides C, et al. Renewable Energy, 2021, 176, 89-105



Application for mooring breakage analysis

<u>Yang Y</u>, Bashir M, Li C, et al. Ocean Engineering, 2021, 233: 108887.

- ✓ F2A cannot accurately capture the nonlinear aeroelastic behavior of long, flexible blades used in large-scale FOWTs
- ✓ Fails for eccentric FOWT concepts, e.g WindFloat
- ✓ Underestimation of the tower's natural frequency without consideration of the hydrostatic stiffness and added mass of the platform

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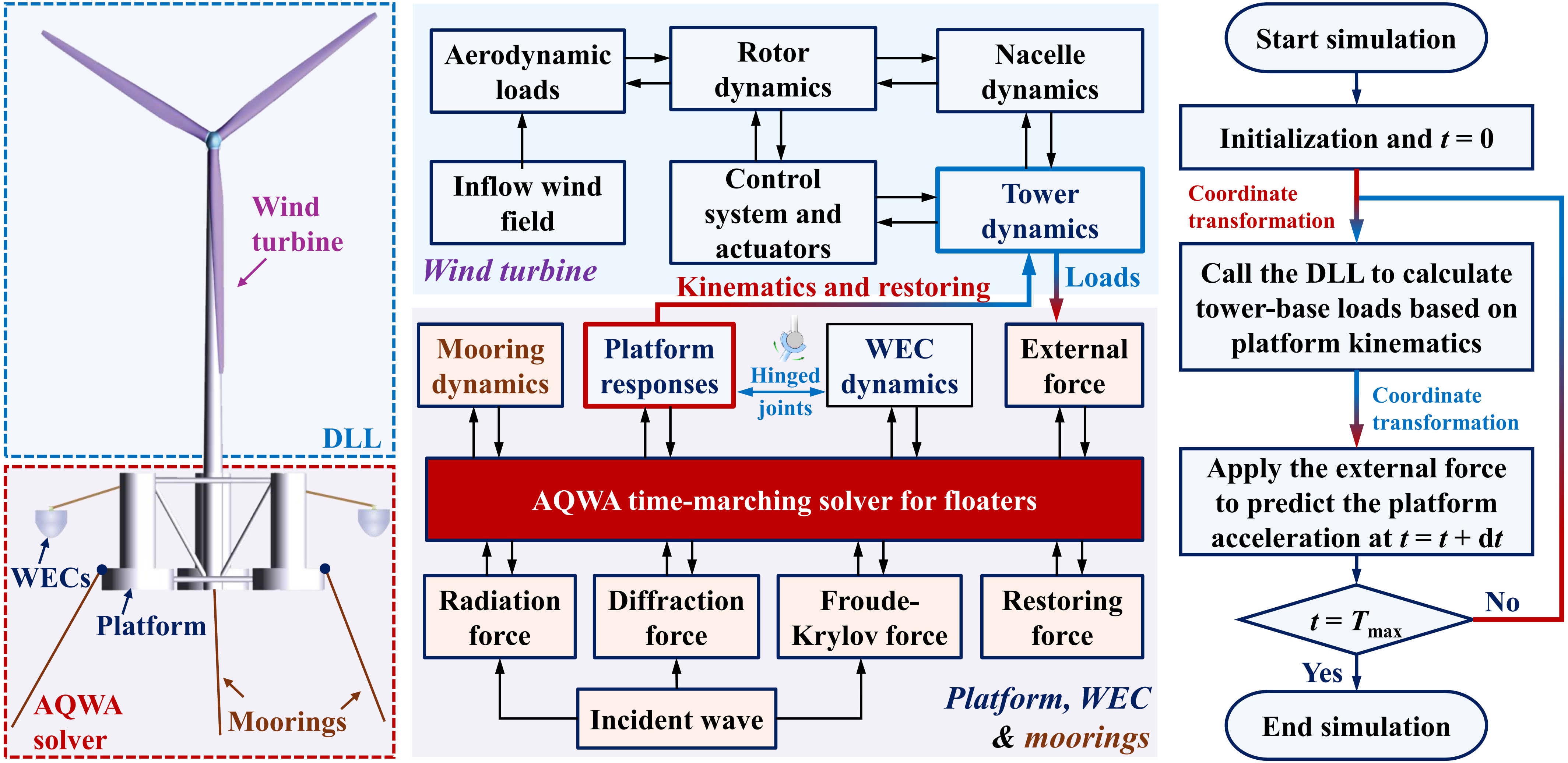
Development

Framework of OpenF2A



Logic diagram of the OpenF2A framework

Transformation of platform motions

Platform motions **output** by the Solver (Displacement, Velocity, Acceleration) → Platform motions **transmit** to the DLL (Displacement, Velocity, Acceleration)

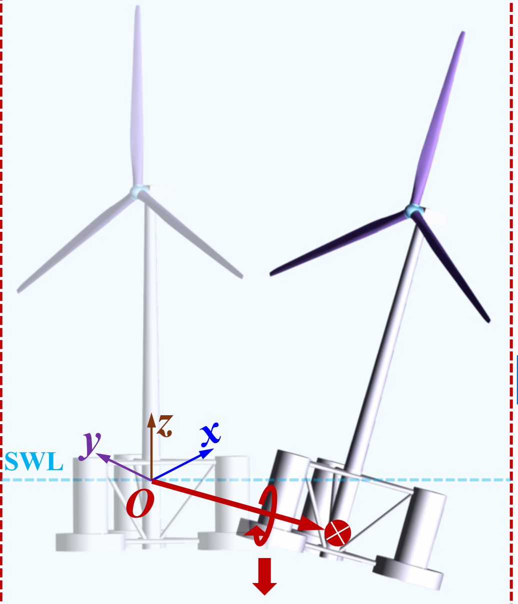
Rigid body motion transformation based on Euler Angle

Translational displacement： $D_{\mathrm{OF}} = D_{\mathrm{AQWA}} - T_{\mathrm{Euler}} \cdot L_{\mathrm{ref}} - P_{OO'}$

Translational velocity： $V_{\mathrm{OF}} = V_{\mathrm{AQWA}} - T_{\mathrm{Euler}} \cdot L_{\mathrm{ref}} \times \omega_{\mathrm{AQWA}}$

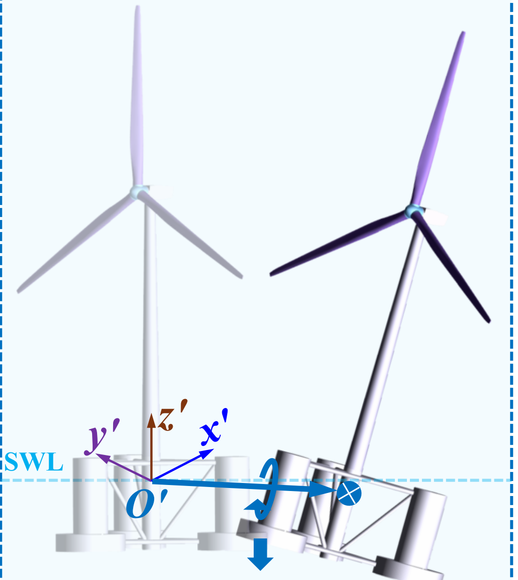
Acceleration： $A_{\mathrm{OF}} = \left(V_{\mathrm{OF}} - V'_{\mathrm{OF}}\right)/\mathrm{d}t$

$$T_{\mathrm{Euler}} = \begin{bmatrix} cy \cdot cz & sx \cdot sy \cdot cz - cx \cdot sz & cx \cdot sy \cdot cz + sx \cdot sz \\ cy \cdot sz & sx \cdot sy \cdot sz + cx \cdot cz & cx \cdot sy \cdot sz - sx \cdot cz \\ -sy & sx \cdot cy & cx \cdot cy \end{bmatrix}$$



$\begin{bmatrix} D_{\mathrm{AQWA}} \\ \theta_{\mathrm{AQWA}} \end{bmatrix}, \begin{bmatrix} V_{\mathrm{AQWA}} \\ \omega_{\mathrm{AQWA}} \end{bmatrix}, \begin{bmatrix} A_{\mathrm{AQWA}} \\ \alpha_{\mathrm{AQWA}} \end{bmatrix}$

Kinematics at CM referred to the global coordinate system O of AQWA



$\begin{bmatrix} D_{\mathrm{OF}} \\ \theta_{\mathrm{OF}} \end{bmatrix}, \begin{bmatrix} V_{\mathrm{OF}} \\ \omega_{\mathrm{OF}} \end{bmatrix}, \begin{bmatrix} A_{\mathrm{OF}} \\ \alpha_{\mathrm{OF}} \end{bmatrix}$

Kinematics at the reference point referred to the global coordinate system O' of OpenFAST

Transformation of tower-base loads

Loads output by the DLL (Force and moment) → **Loads transmit to the Solver (Force and moment)**

Tower-base or local coordinate system → **CM or global coordinate system**

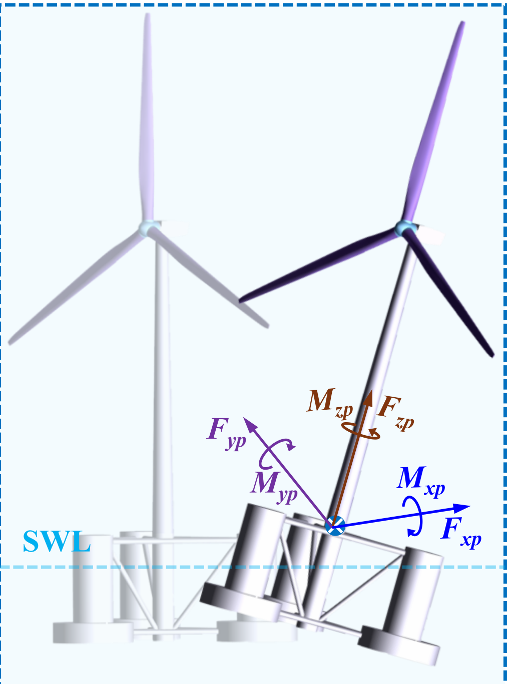
Rigid body motion transformation based on Euler Angle

Translational loads: $$F = T_{\text{Euler}} \cdot F_{\text{p}}$$

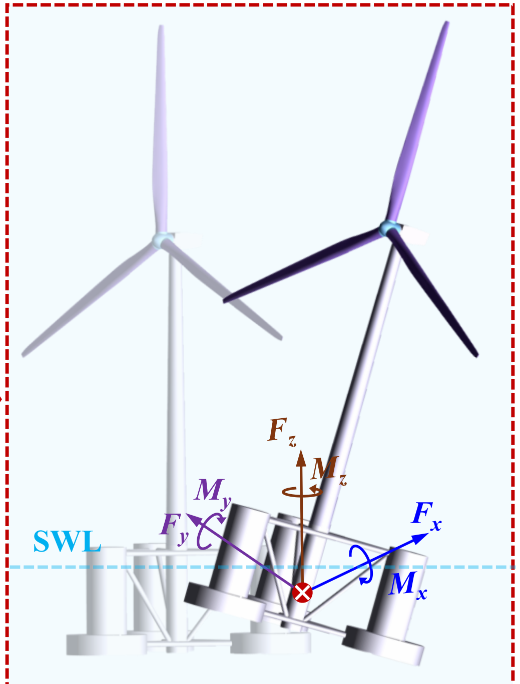
Rotating loads: $$M = T_{\text{Euler}}^{-1} \cdot \left[M_{\text{p}} + \left(L_{\text{ref}} + P_{OO'} \right) \times F_{\text{p}} \right]$$

Tower-base → CM

Local coordinate system → Global coordinate system



Tower base loads referred to the local platform coordinate system



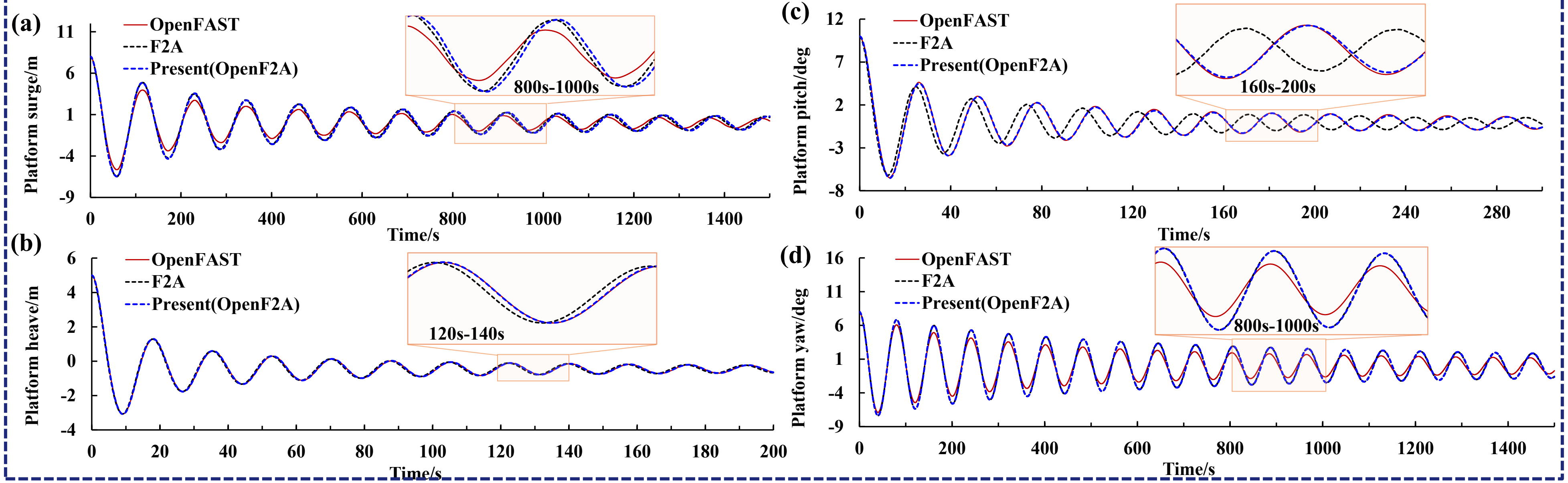
Platform loads at CM referred to the global coordinate system

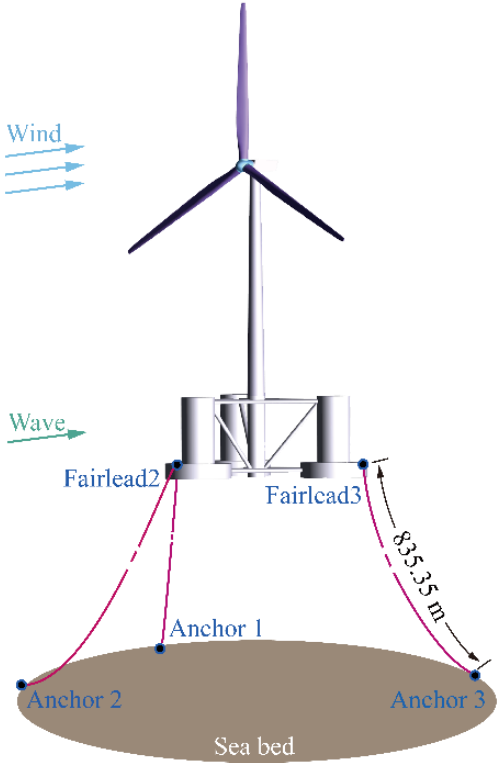
3

Model validations

5MW DeepCWind model

Free decay analysis of platform motions





DeepCWind 5MW wind turbine model

Comparisons of natural periods of the FOWT (Unit: s)

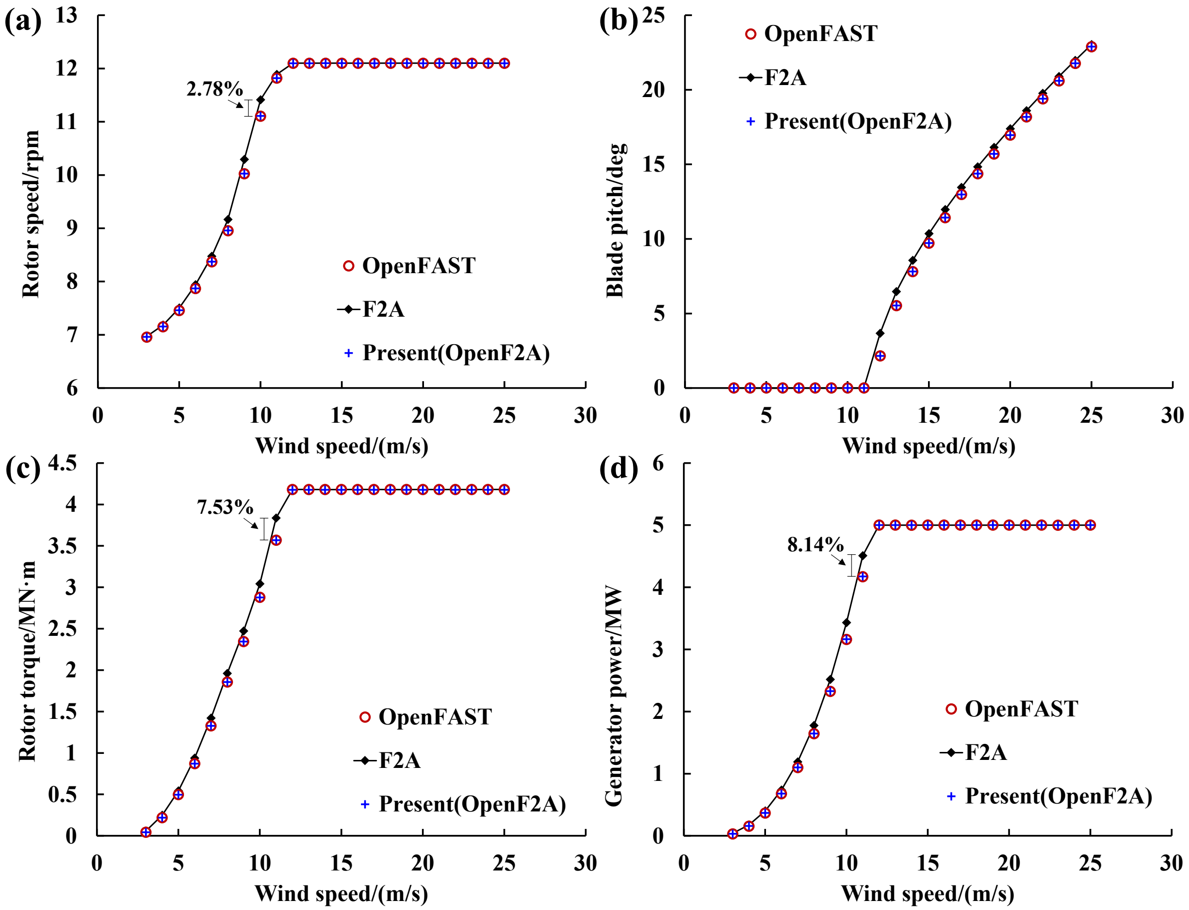
	OpenFAST	F2A	OpenF2A
Surge	114.14	114.46	115.01
Heave	17.56	17.46	17.54
Pitch	**25.86**	**24.47**	**25.87**
Yaw	80.45	80.62	80.67

The results obtained using OpenF2A framework agree well with the predictions of OpenFAST.

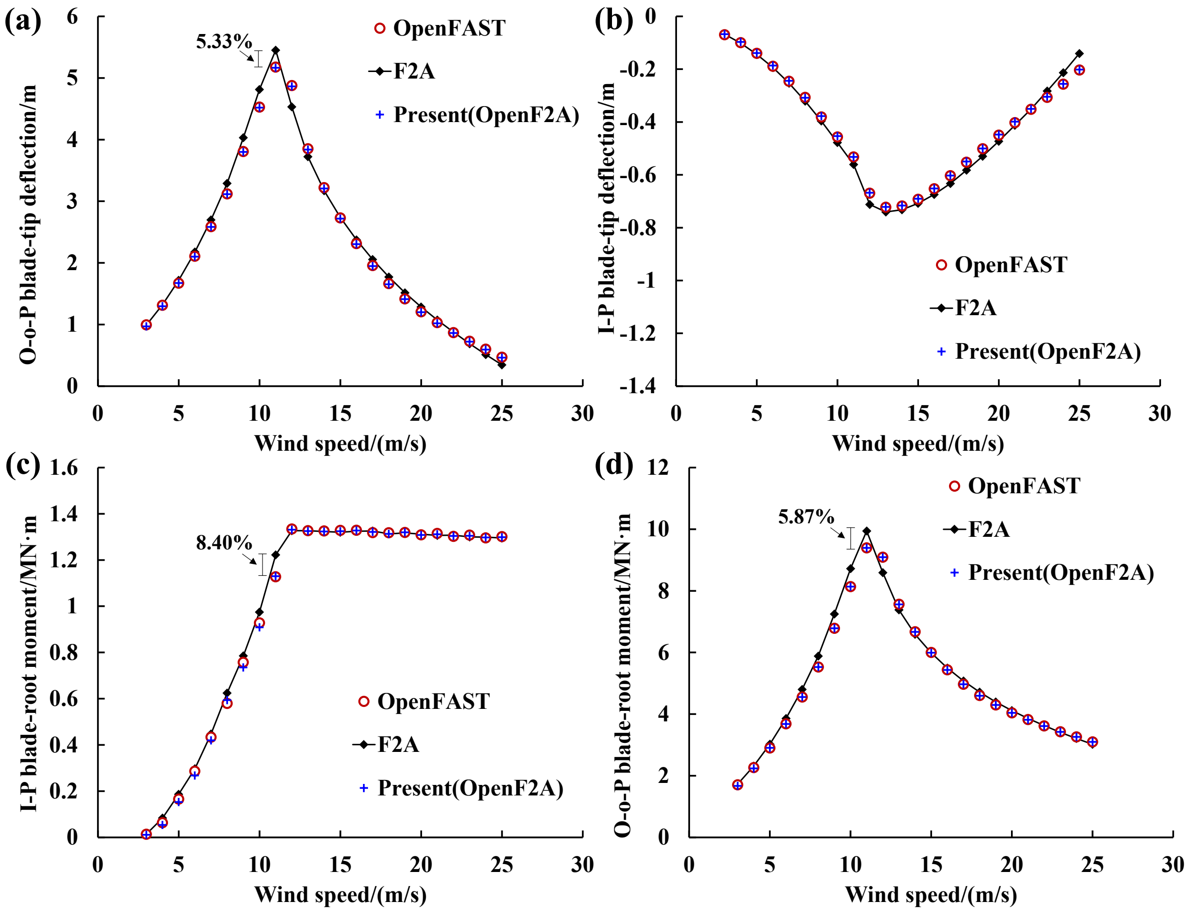
OpenF2A is improved by retaining the terms associated with platform DOFs in the wind turbine's equation of motion, compared to previous F2A.

5MW DeepCWind model

Drivetrain and control system



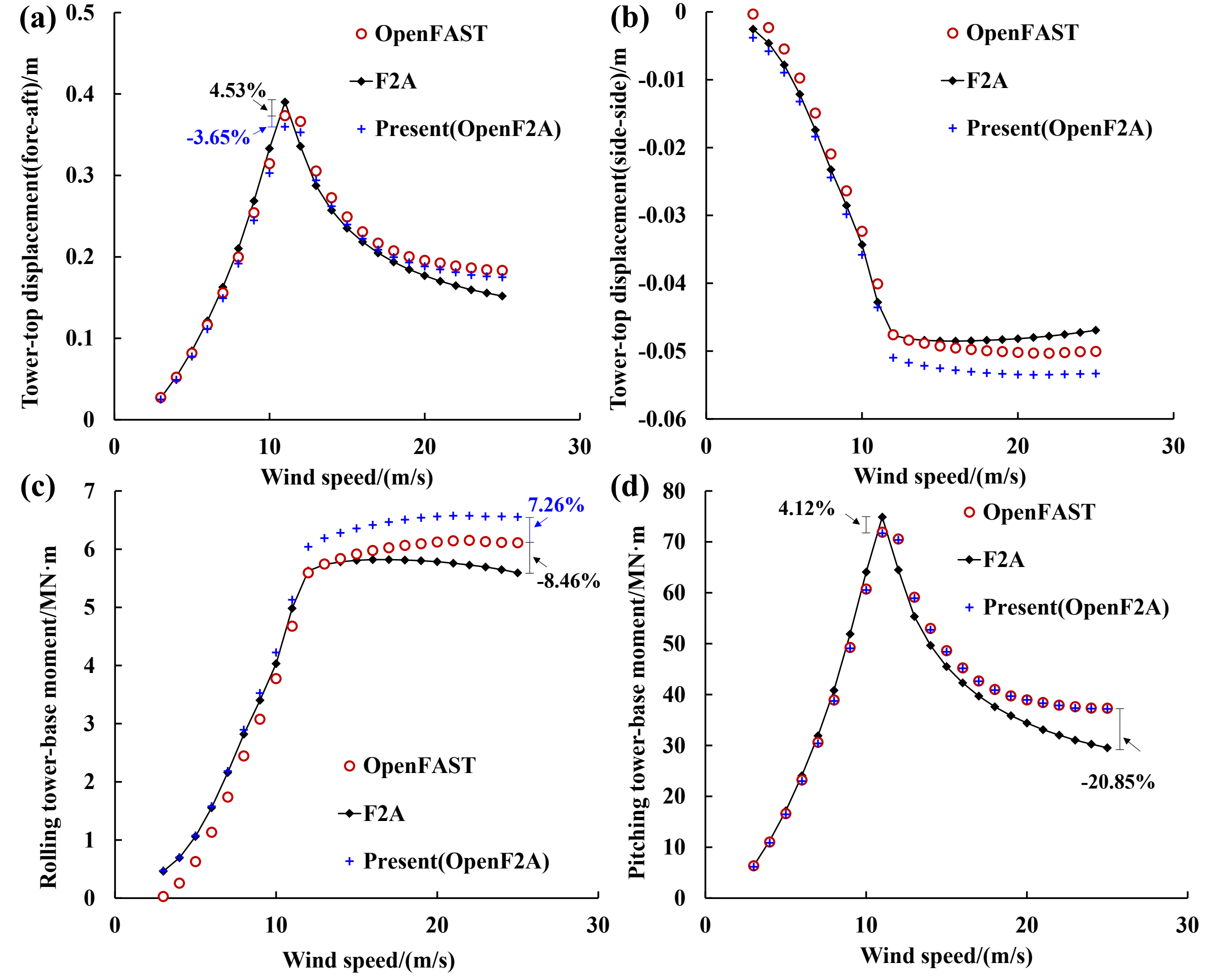
Blade-tip deflections/blade-root bending moments



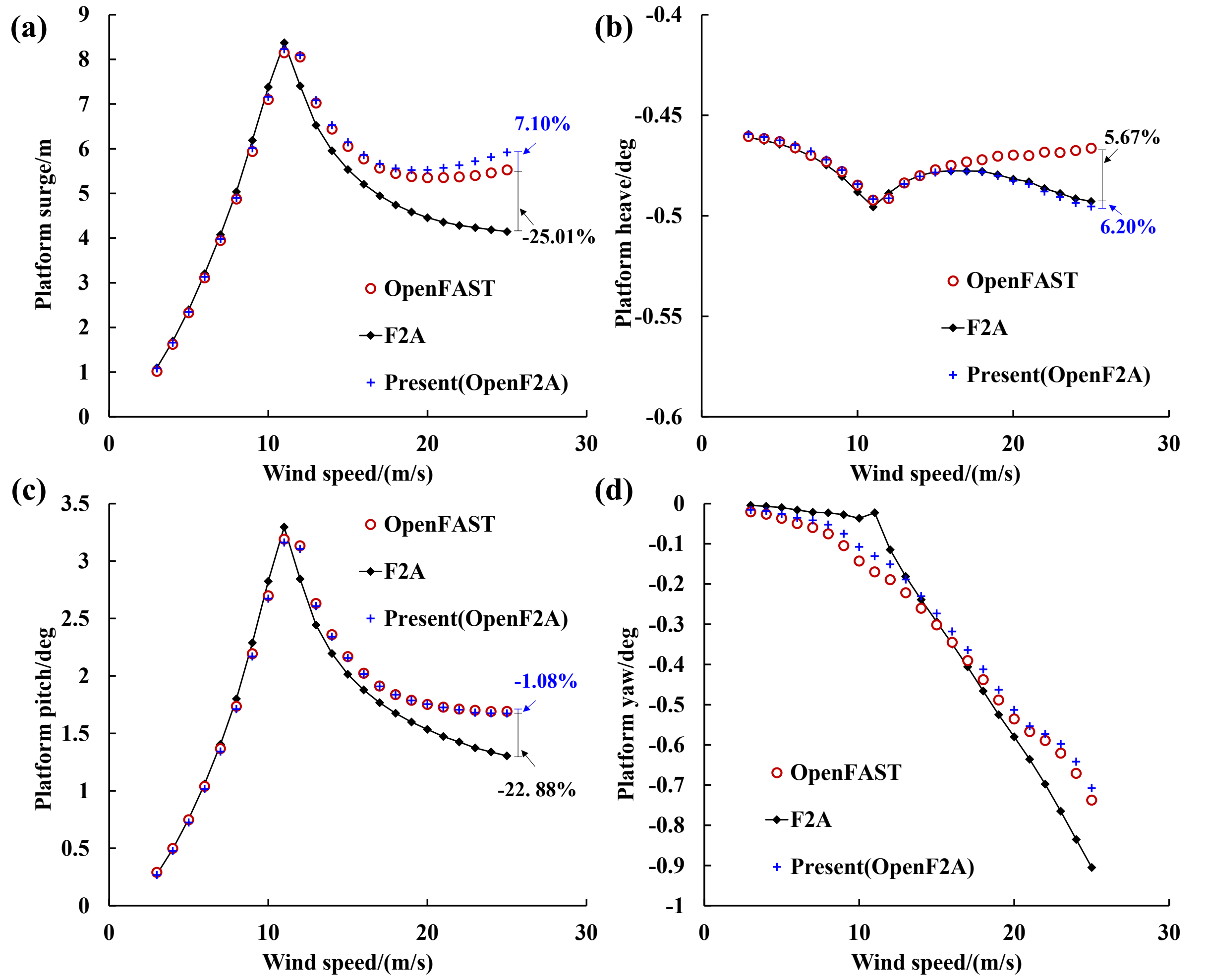
1. Backgroud

5MW DeepCWind model

Tower-top displacements/ Tower-base bending moments

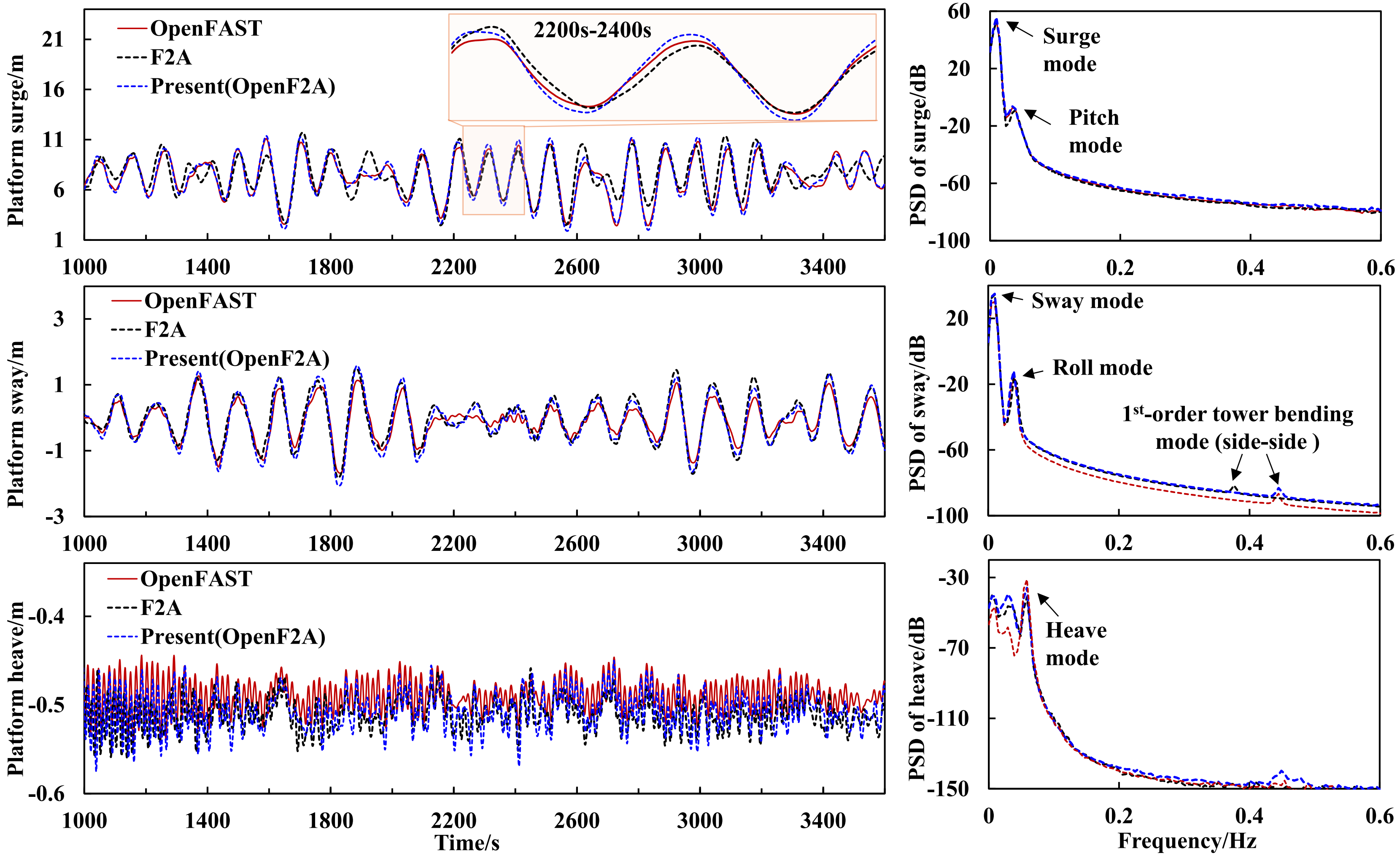


Platform motions



5MW DeepCWind model

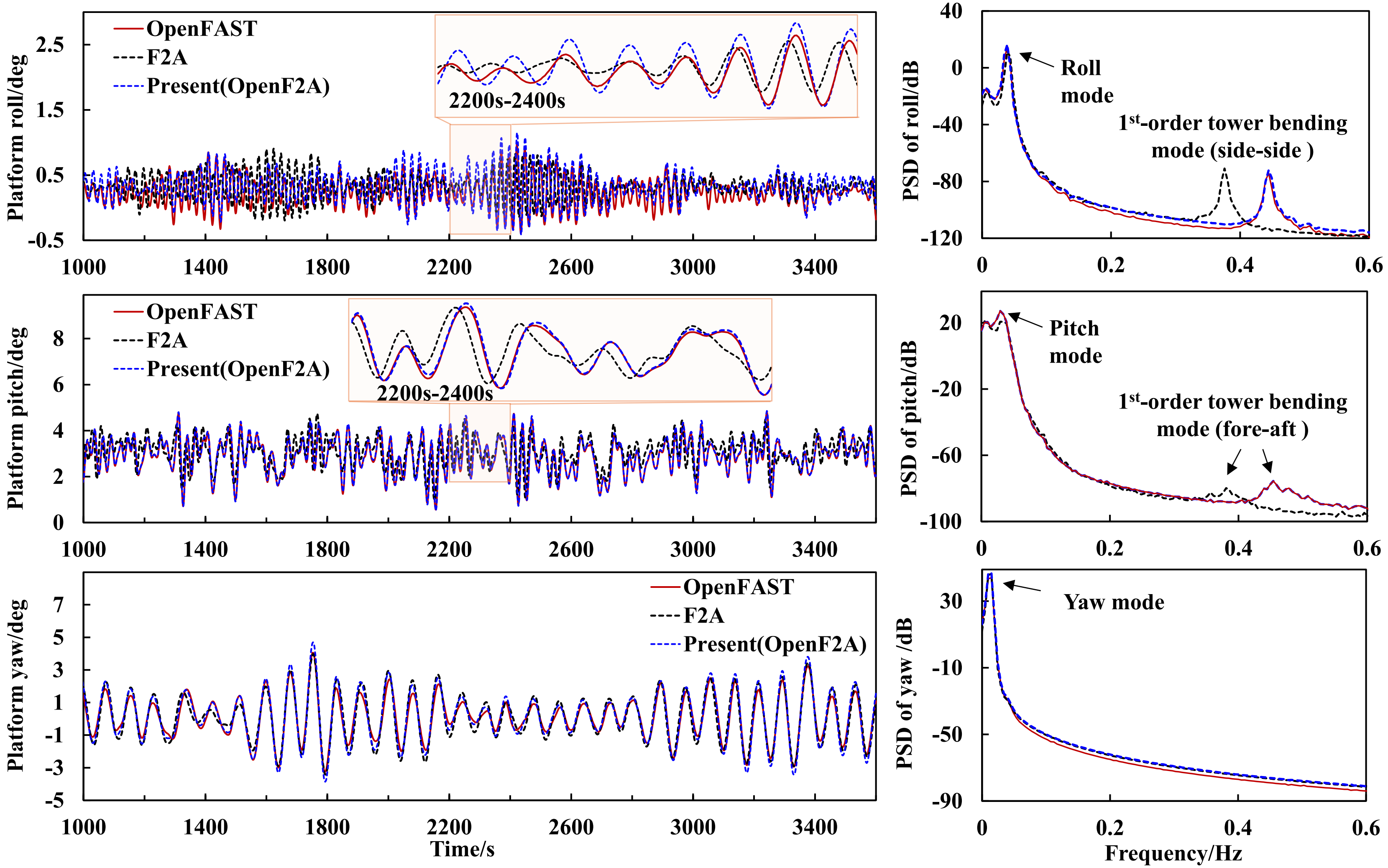
Translational platform motions and the corresponding PSD distribution



- The variation trend of platform surge and sway predicted by OpenF2A aligns with that calculated by OpenFAST.
- The comparison of PSD distributions indicates that discrepancies of heave motion are mainly attributed to the surge and heave modes.

5MW DeepCWind model

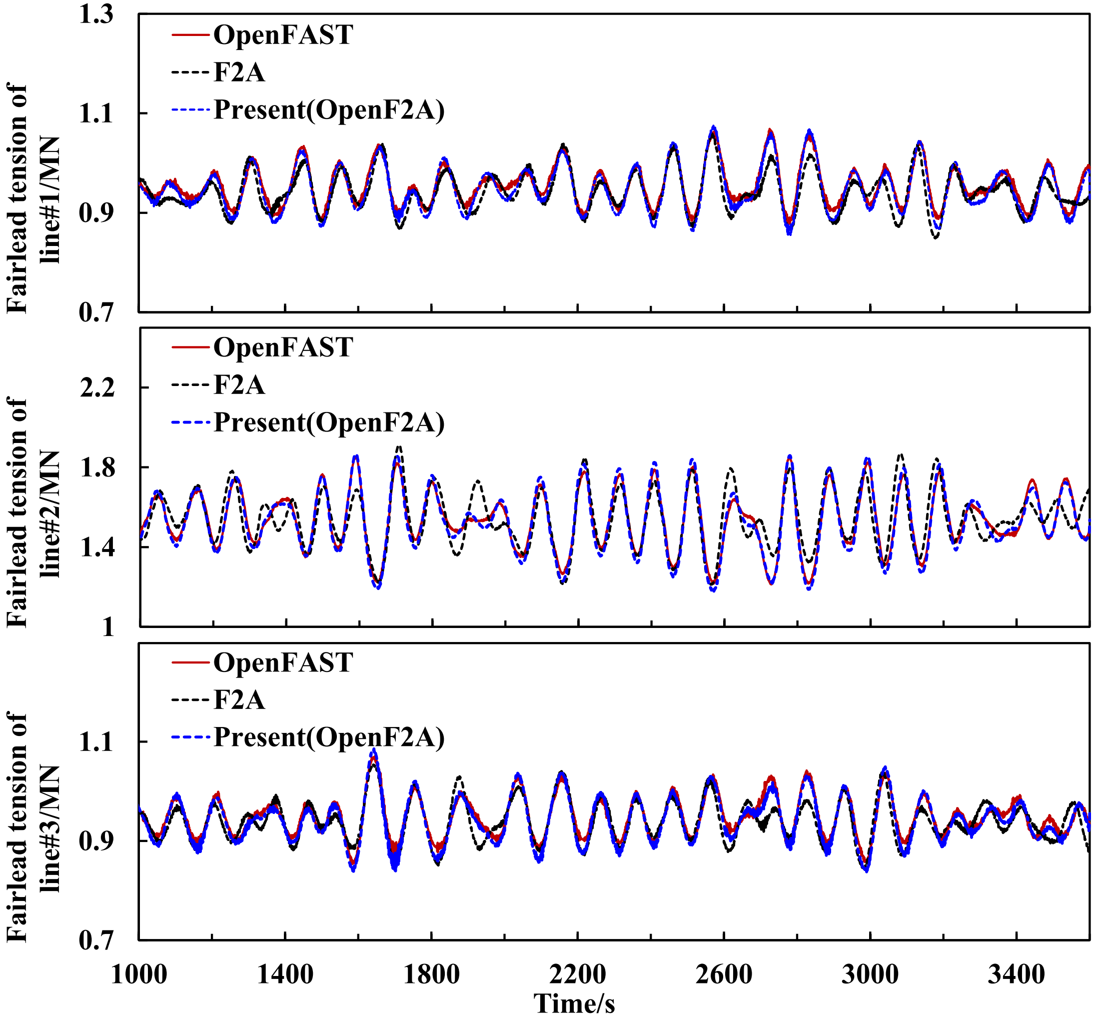
Angular platform motions and the corresponding PSD distribution



- Good agreements are obtained between the pitch motions calculated by OpenF2A and OpenFAST, although slight difference in time variations is observed.
- OpenF2A is capable of producing reliable platform responses of FOWTs.

5MW DeepCWind model

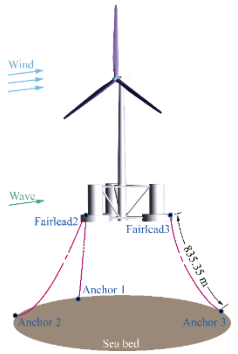
Fairlead tension in mooring lines



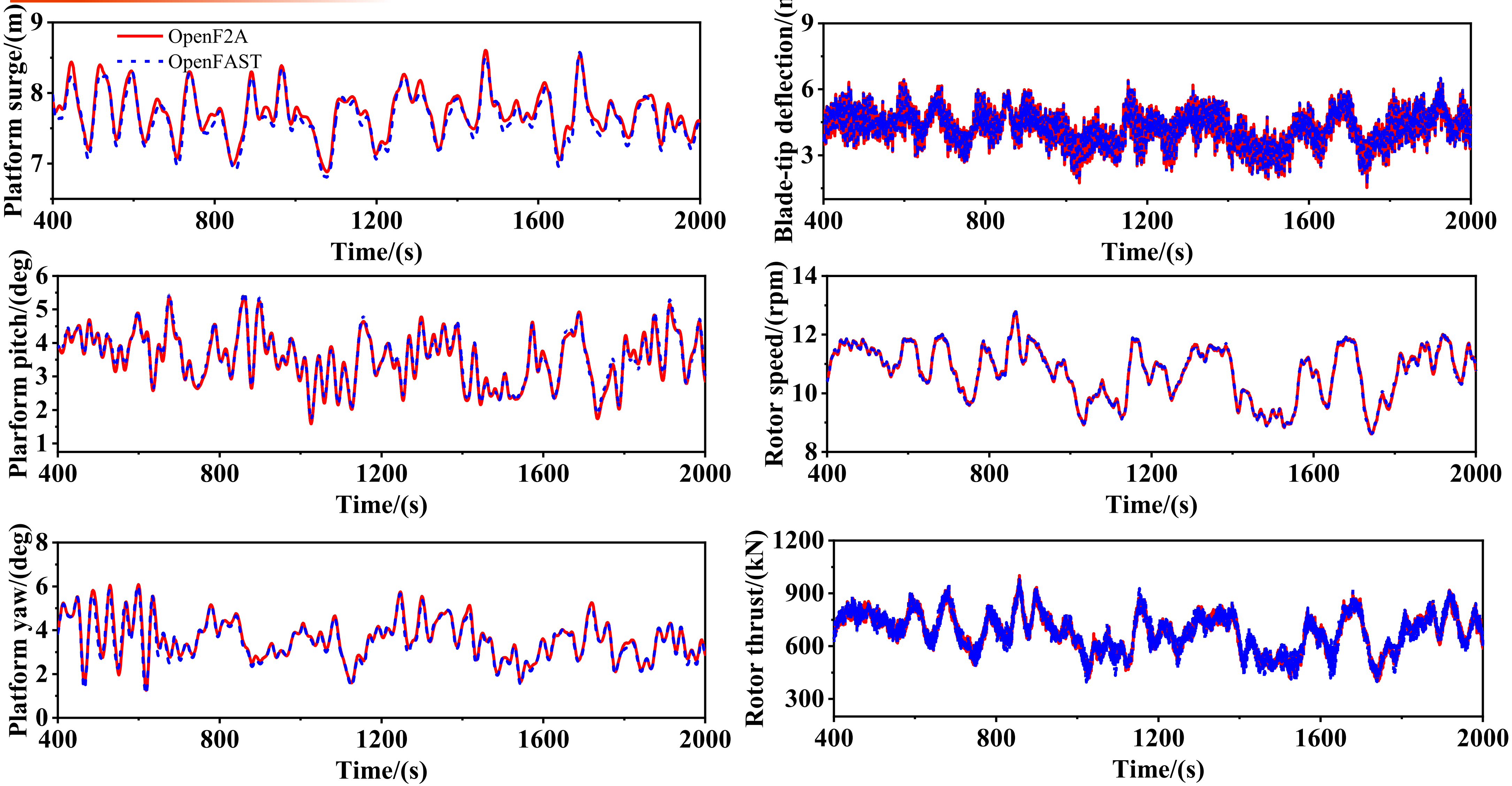
Statistical tension in the mooring lines

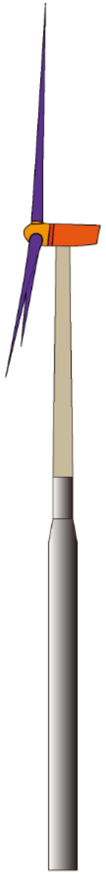
		OpenFAST	F2A		Present (OpenF2A)	
		Value (MN)	Value (MN)	Difference	Value (MN)	Difference
Line#1	Mean	0.955	0.942	1.34%	0.947	0.75%
	Min	0.874	0.849	2.94%	0.853	2.41%
	Max	1.070	1.060	0.98%	1.077	-0.57%
Line#2	Mean	1.537	1.541	-0.32%	1.530	0.46%
	Min	1.213	1.212	0.06%	1.177	2.92%
	Max	1.855	1.911	-3.04%	1.869	-0.78%
Line#3	Mean	0.949	0.940	0.96%	0.942	0.70%
	Min	0.851	0.844	0.87%	0.835	1.86%
	Max	1.073	1.054	1.77%	1.087	-1.30%

The discrepancies of mooring tension predicted by OpenF2A compared to OpenFAST and F2A are less than 4%.



5MW OC3Hywind model

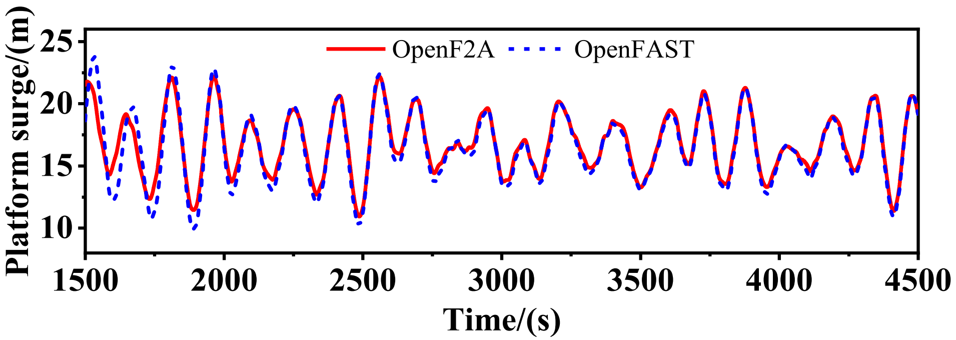


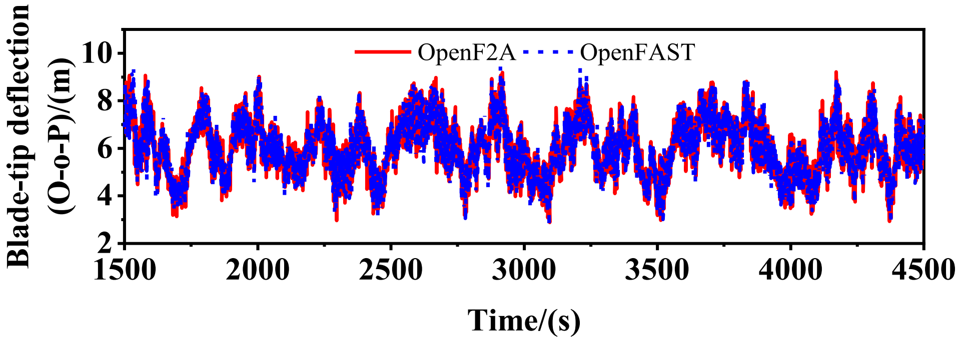


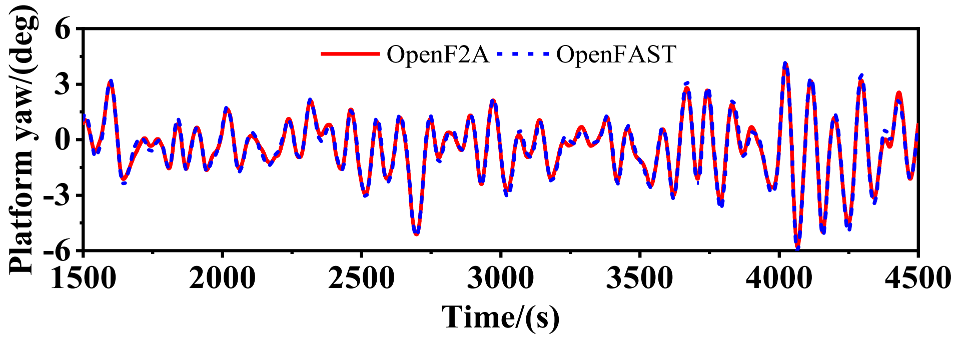
5MW
OC3Spar

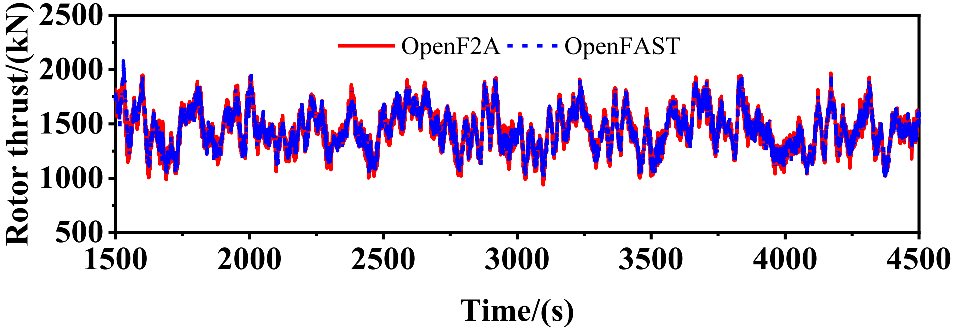
3. Model validation

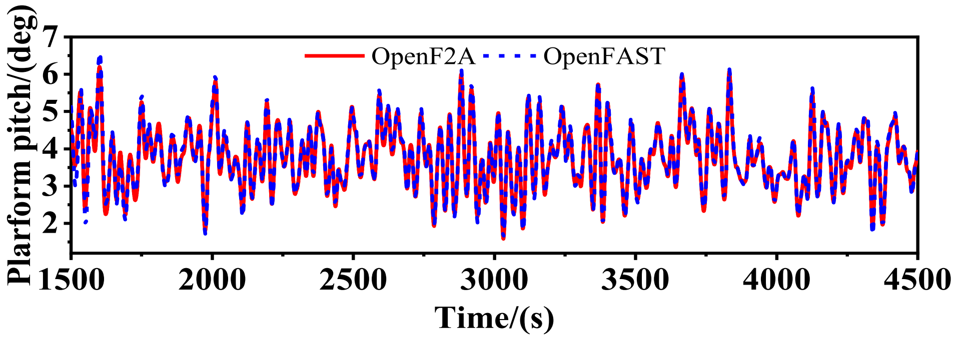
10MW OOStar model

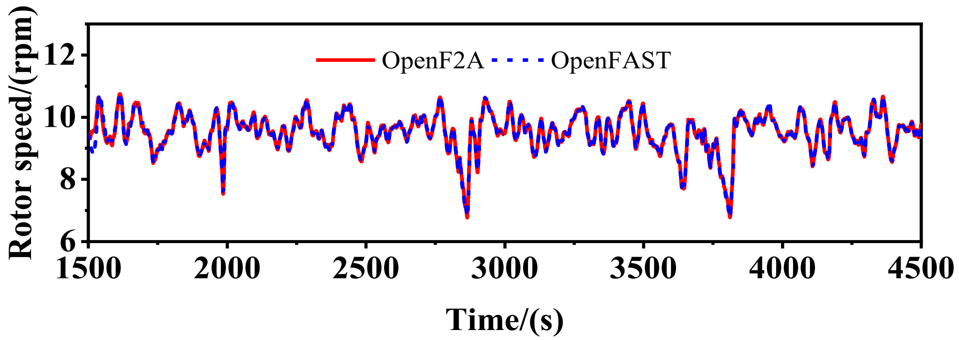


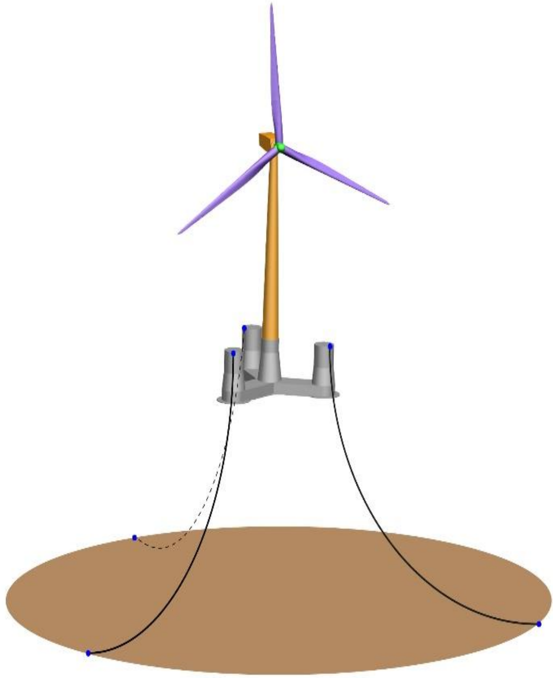






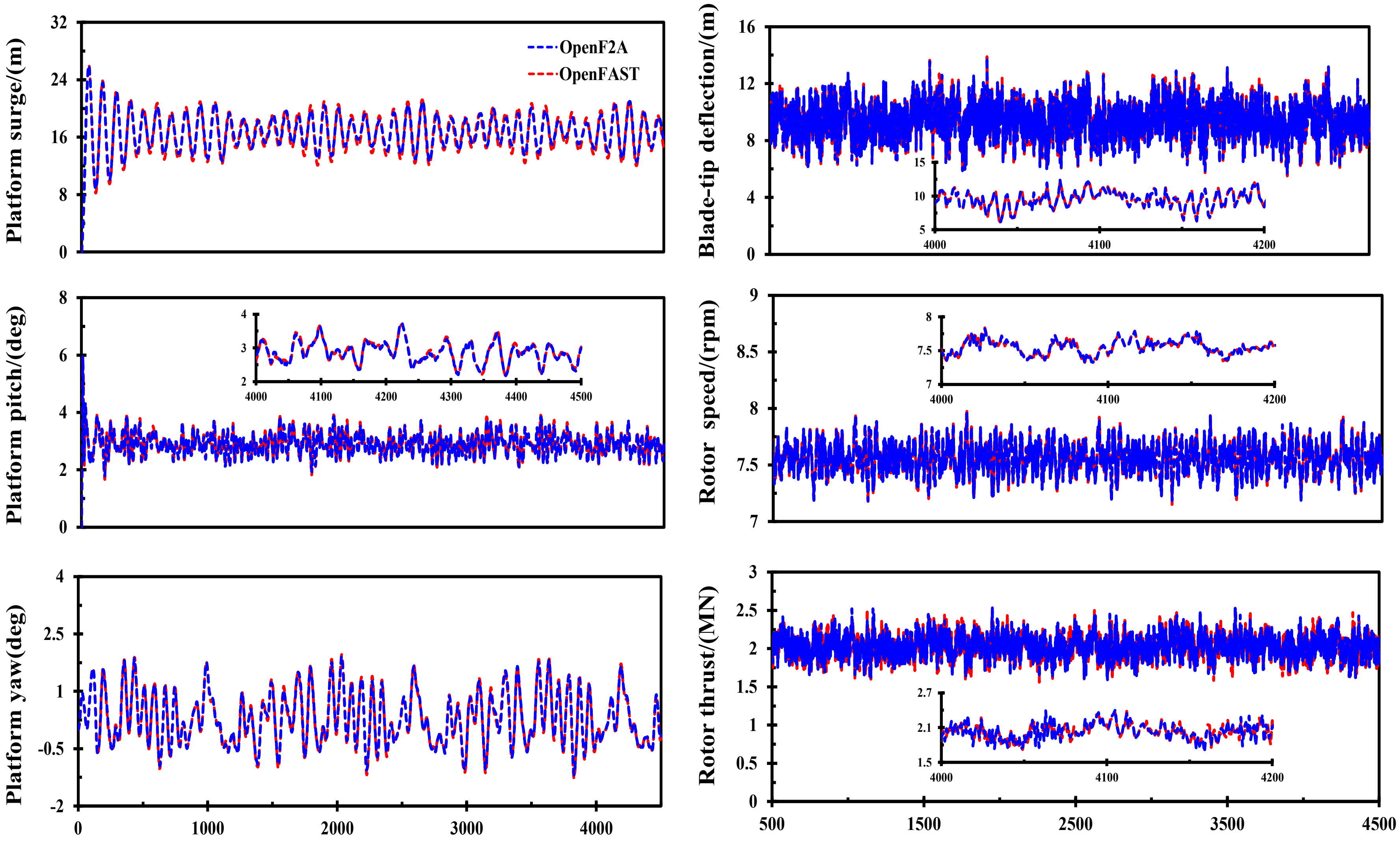






10MW
OOStar model

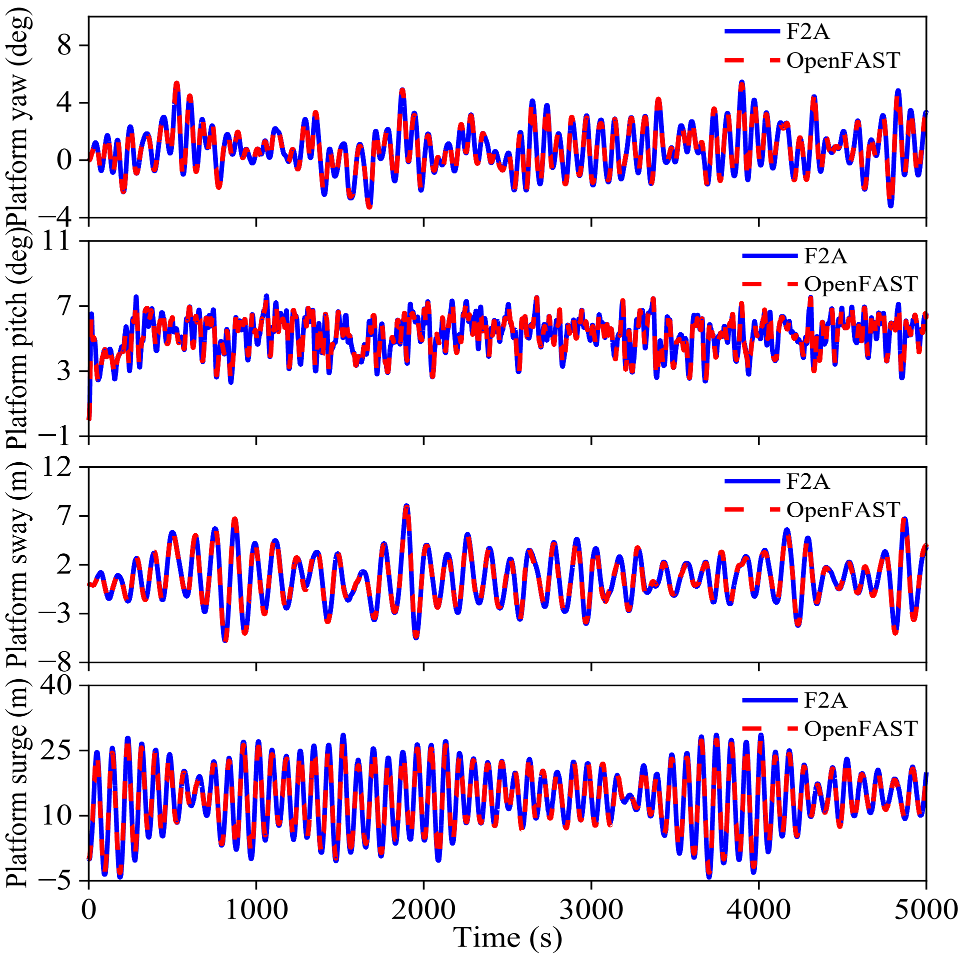
15MW VolturnUS-S model

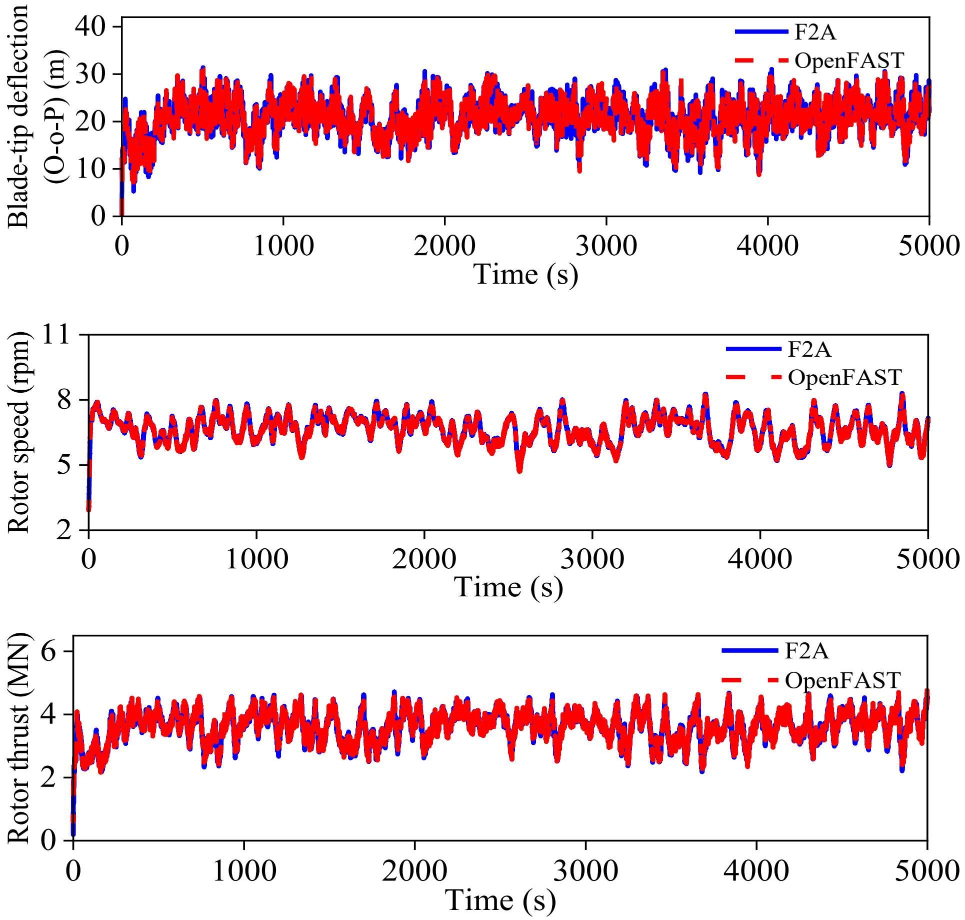


15MW VolturnUS-S model

3. Model validation

IEA22MW VolturnUS-S model





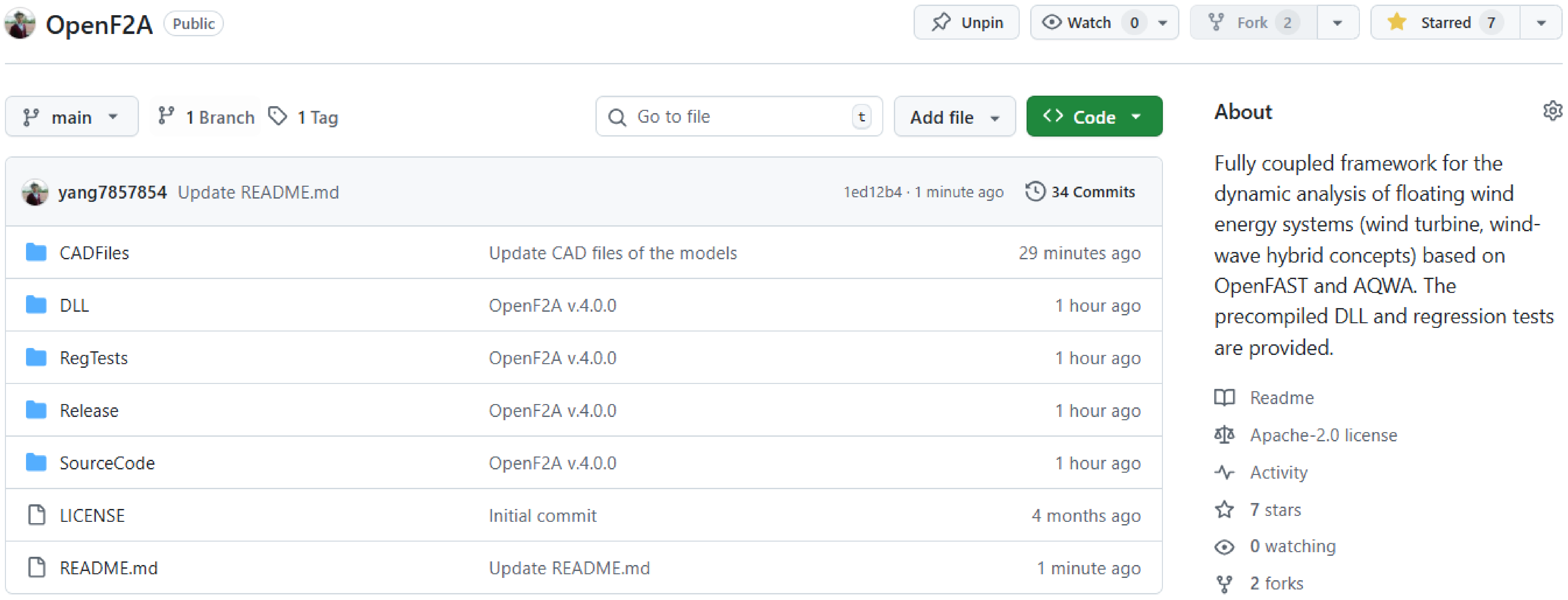
IEA22MW VolturnUS-S model(scaled up based on 15MW VolturnUS)

4

User guide

Open source files

OpenF2A is released on Github, which can accessed by: https://github.com/yang7857854/OpenF2A/tree/main



DLL files

OpenF2A is actually a DLL that is invoked by AQWA to calculate external loads on the floating bodies.

Release https://github.com/yang7857854/OpenF2A/releases/tag/OpenF2A_v4.0.0

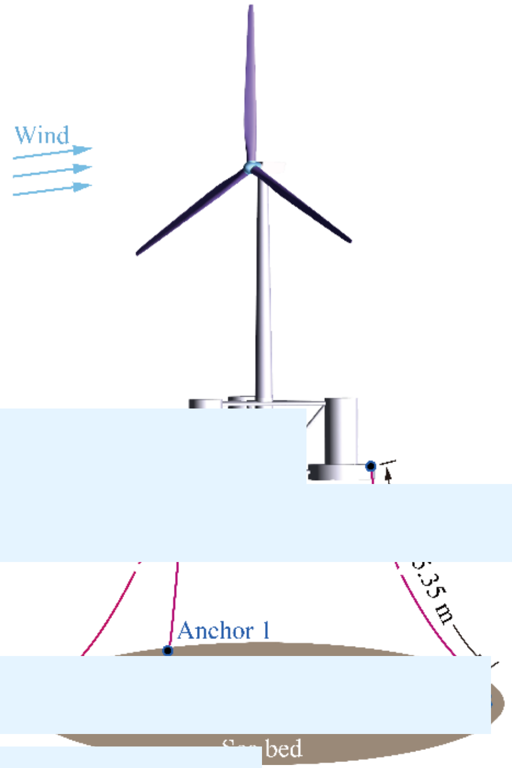
Assets 9

InputforOpenF2A.txt	sha256:db8d4fff82d1451e24c98cf4e1750…	1.42 KB	34 minutes ago
InputforreadAQWAResults_SingleFile.txt	sha256:26e0346757bdd2139291ef57a4a52…	1.73 KB	34 minutes ago
openfast_x64_4.0.exe	sha256:3990185812a7b8d8021da9f5fe0ee…	50 MB	33 minutes ago
readAQWAResults_SingleFile.exe	sha256:2c451809f327057aa37c28f80f273…	68 KB	34 minutes ago
TurbSim_x64.exe	sha256:7f70daa00a876b46c495acdb16f26…	25.7 MB	34 minutes ago
user_force64-OpenF2A-4.0-option.dll	sha256:9f075e37902de165efd9243ad8322…	46 MB	34 minutes ago
user_force64-OpenF2A-v4.0.0.dll	sha256:837c6301fdeec98844ebdeb66300d…	46 MB	34 minutes ago
Source code (zip)			Nov 18, 2025
Source code (tar.gz)			Nov 18, 2025

4. User guide

Use OpenF2A

Case study of the 5MW DeepCWind model



Step1: Download essential files

Assets 9

- InputforOpenF2A.txt — Example input of OpenF2A
- InputforreadAQWAResults_SingleFile.txt — Input for the result extraction code
- openfast_x64_4.0.exe
- readAQWAResults_SingleFile.exe — Extract results of AQWA
- TurbSim_x64.exe — TurbSim to generate wind data
- user_force64-OpenF2A-4.0-option.dll
- user_force64-OpenF2A-v4.0.0.dll — DLL to be used
- Source code (zip) — Examples, source code and other files given in the main repository
- Source code (tar.gz)

Use OpenF2A

Step2: rename, copy and replace the userforce DLL

Rename it as “user_force64.dll”, copy it to the path of AQWA bin files, replace the DLL

The bin file path in my computer is: D:\Program Files\ANSYS Inc\v241\aqwa\bin\winx64

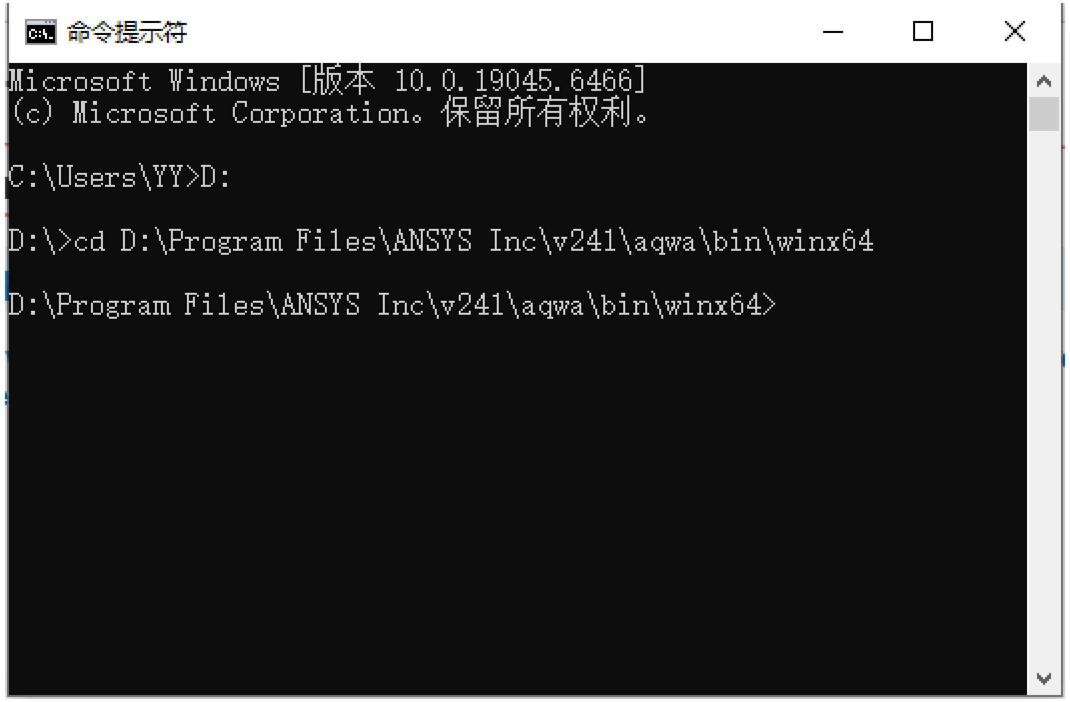
Program Files › ANSYS Inc › v241 › aqwa › bin › winx64

名称	修改日期	类型	大小
user_force64.dll	3/9/2026 12:44 AM	应用程序扩展	47,085 KB
AqwaWave.exe	11/6/2023 8:24 PM	应用程序	96,232 KB
Ags.exe	11/6/2023 8:24 PM	应用程序	147,691 KB
Aqwa.exe	11/6/2023 8:24 PM	应用程序	141,139 KB
Aqwa2Neut.exe	11/6/2023 8:24 PM	应用程序	84,804 KB
AqwaFlow.exe	11/6/2023 8:24 PM	应用程序	84,799 KB
wdrvr.dll	11/6/2023 8:24 PM	应用程序扩展	80 KB
ApipWrapper.dll	11/6/2023 8:03 PM	应用程序扩展	957 KB
libcrypto-1_1-x64.dll	10/5/2023 7:38 PM	应用程序扩展	2,781 KB
libssl-1_1-x64.dll	10/5/2023 7:38 PM	应用程序扩展	669 KB
ag.tin	8/29/2023 1:48 PM	ANSYS 2020 R1	1,947 KB
zlib1.dll	8/2/2023 6:27 PM	应用程序扩展	130 KB

Use OpenF2A

Step3: open command promote window, change path to AQWA bin files

(1) Press <win> + <R>, type “cmd”, or click ‘start’, type “CMD” to open the command promote window
(2) Change directory to the AQWA bin folder using “cd” command



On my computer, the AQWA bin path is located in the D drive. We need to change the drive index, and then change directory.

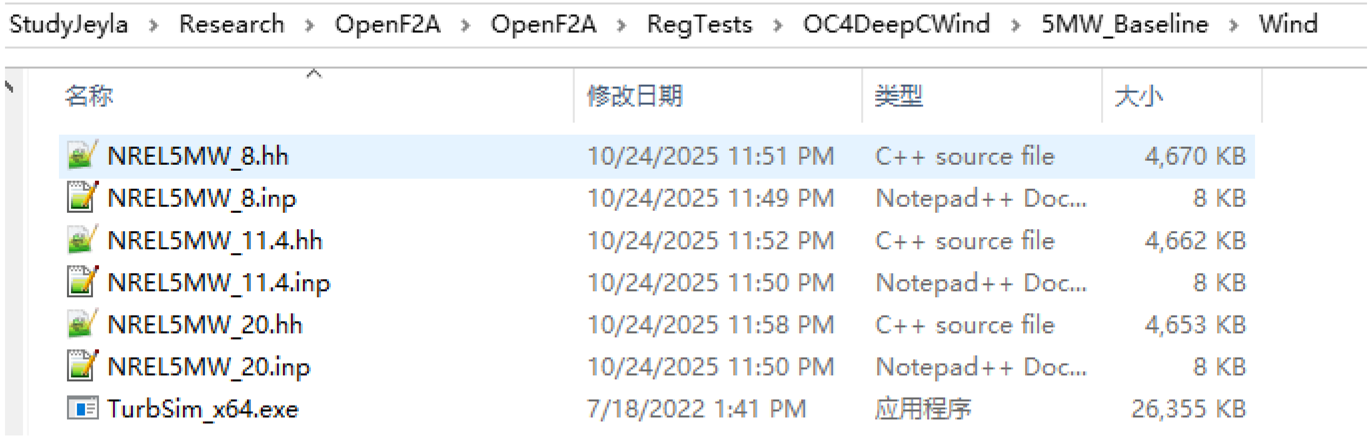
a) Type *D:*
b) Press *Enter*
c) Type *cd D:\Program Files\ANSYS Inc\v241\aqwa\bin\winx64*
d) Press *Enter*

Use OpenF2A

Step4: Generate wind data

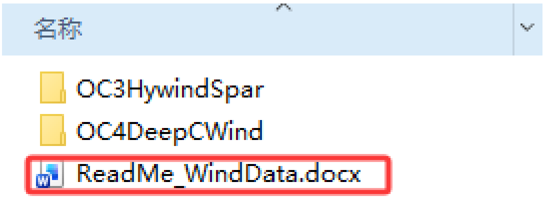
Copy the TurbSim program downloaded previously to the path of wind data that is:

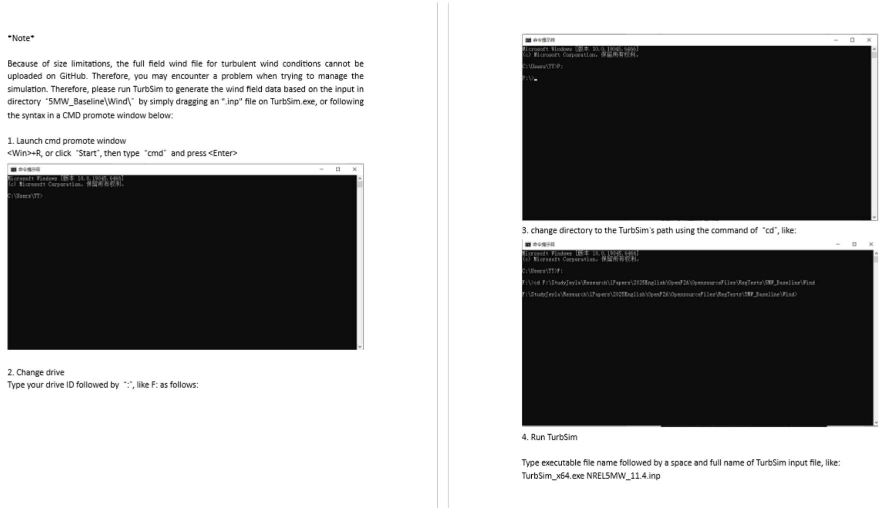
*/OpenF2A/RegTests/OC4DeepCWind/5MW_Baseline/Wind/.



名称	修改日期	类型	大小
NREL5MW_8.hh	10/24/2025 11:51 PM	C++ source file	4,670 KB
NREL5MW_8.inp	10/24/2025 11:49 PM	Notepad++ Doc...	8 KB
NREL5MW_11.4.hh	10/24/2025 11:52 PM	C++ source file	4,662 KB
NREL5MW_11.4.inp	10/24/2025 11:50 PM	Notepad++ Doc...	8 KB
NREL5MW_20.hh	10/24/2025 11:58 PM	C++ source file	4,653 KB
NREL5MW_20.inp	10/24/2025 11:50 PM	Notepad++ Doc...	8 KB
TurbSim_x64.exe	7/18/2022 1:41 PM	应用程序	26,355 KB

Run TurbSim to generate the wind data, following the description in the document: ReadMe_WindData.docx located in “RegTests”





Note

Because of size limitations, the full field wind file for turbulent wind conditions cannot be uploaded on GitHub. Therefore, you may encounter a problem when trying to manage the simulation. Therefore, please run TurbSim to generate the wind field data based on the input in directory “5MW_Baseline\Wind\” by simply dragging an ".inp" file on TurbSim.exe, or following the syntax in a CMD promote window below:

1. Launch cmd promote window
<Win>+R, or click “Start”, then type “cmd” and press <Enter>

2. Change drive
Type your drive ID followed by “:”, like F: as follows:

3. change directory to the TurbSim's path using the command of “cd”, like:

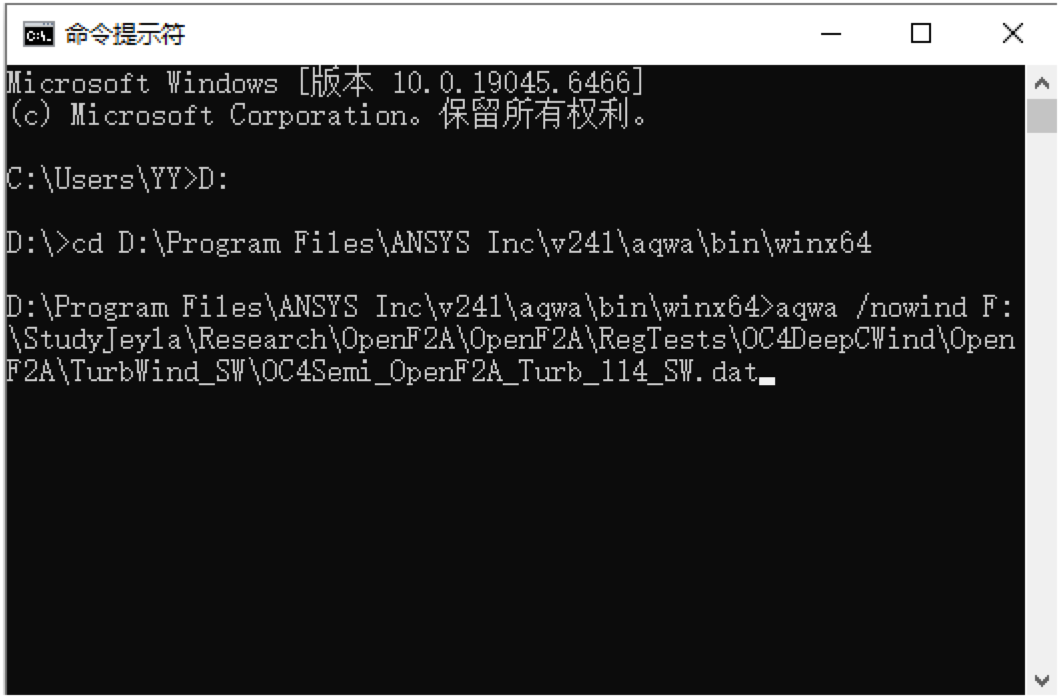
4. Run TurbSim

Type executable file name followed by a space and full name of TurbSim input file, like:
TurbSim_x64.exe NREL5MW_11.4.inp

Use OpenF2A

Step5: Run AQWA program

Run AQWA program following the syntax: aqwa /nowind <full path of AQWA case input file>



aqwa: denotes the program name of AQWA

/nowind: denotes not to displace the progress window of AQWA case

<full path of AQWA case input file>: denotes the full path of the AQWA case input file.

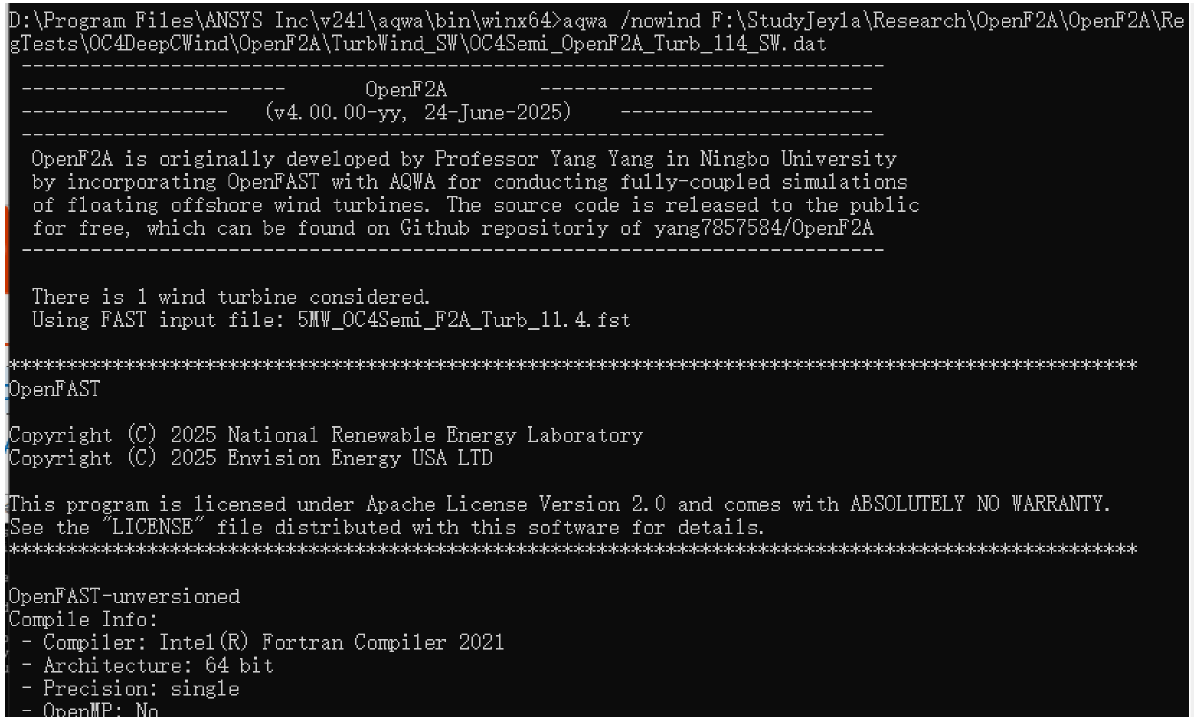
Press <Enter> to run the program

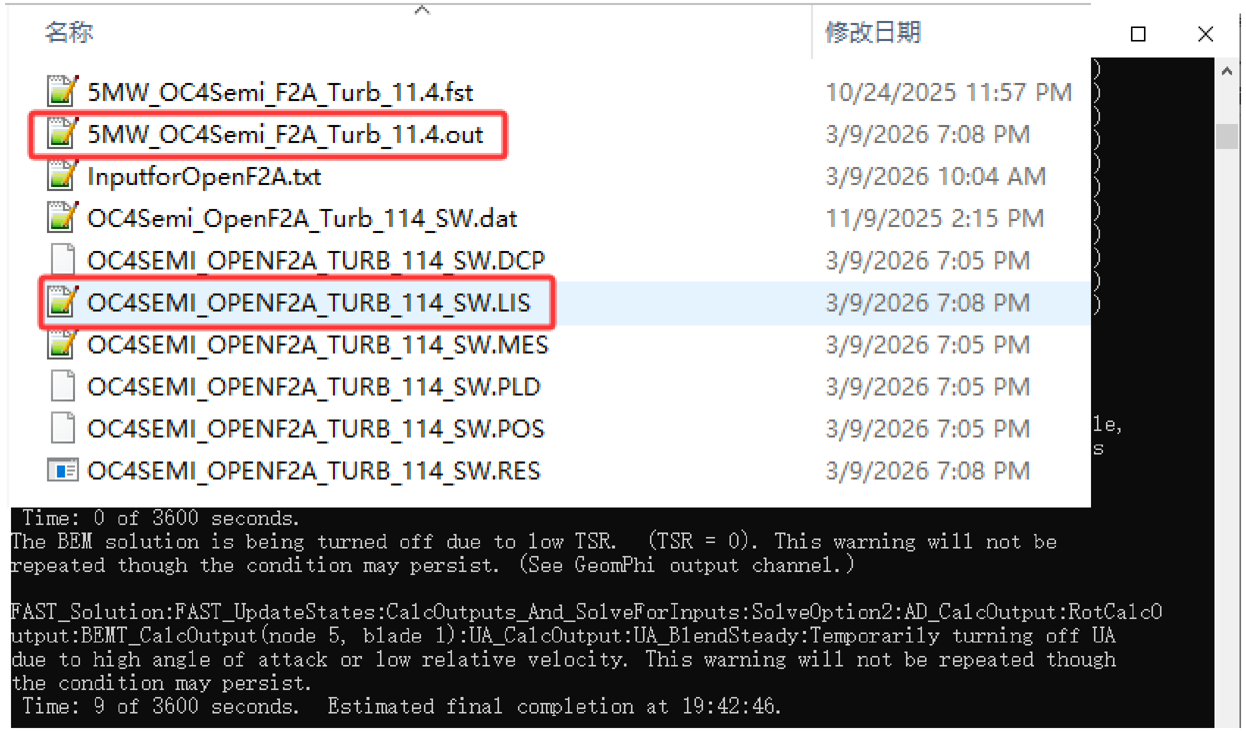
The full path of the case file on my computer is:
F:\StudyJeyla\Research\OpenF2A\OpenF2A\RegTests\OC4DeepCWind\OpenF2A\TurbWind_SW\OC4Semi_OpenF2A_Turb_114_SW.dat

Use OpenF2A

Step6: Generate results

The notice of OpenF2A and OpenFAST subroutine output information will be displayed. The results stored in the <*.out> and <*.lis> files at anytime you want.





Use OpenF2A

Step7: Check results

The *.out file is generated by OpenFAST. The *.lis is generated by AQWA, in which the results can be extracted using “readAQWAResults_SingleFile.exe” downloaded previously.

```

Predictions were generated on 09-Mar-2026 at 19:08:29 using OpenFAST, compiled on Mar 07 2026 at 15:18:45 as
 linked with  NWTC Subroutine Library; ElastoDyn; InflowWind; AeroDyn; ServoDyn

Description from the FAST input file: FAST Certification Test #25: NREL 5.0 MW Baseline Wind Turbine with OC4

Time      Wind1VelX     Wind1VelY     Wind1VelZ     Azimuth RotSpeed     GenSpeed      OoPDefl1      IPDefl1 TwstDefl1     B
(s)  (m/s)     (m/s)     (m/s)      (deg)     (rpm)     (rpm)     (m)  (m)  (deg)     (deg)     (m)  (m)  (deg)     (m)  (m)  (m)  (deg)
     0.0000     1.0824145E+01     1.2700728E+00    -3.0534160E-01     0.0000000E+00     0.0000000E+00     0.0000000E+00
     0.1000     1.0201708E+01     1.2365046E+00    -8.1741667E-01     2.3222642E-03     4.4536660E-03     2.6371709E-01
     0.2000     1.0552862E+01     1.2987436E+00     2.5282921E-02     3.5516983E-03     1.7135729E-03     5.9428090E-01
     0.3000     1.1047563E+01     4.2382103E-01    -2.5953192E-01     6.2291324E-03     6.2702359E-03     4.3719360E-01
     0.4000     1.1118747E+01     4.5984206E-01    -3.1868184E-01     1.0846340E-02     1.0161970E-02     5.5948824E-01
     0.5000     1.0882587E+01     8.1963360E-01    -3.0889821E-01     1.7703850E-02     1.2183229E-02     1.2512579E+00
     0.6000     1.0259258E+01     1.3212348E+00    -8.3663058E-01     2.8249661E-02     2.4832888E-02     2.0298223E+00
     0.7000     9.8868246E+00     3.5674921E-01    -4.8848635E-01     4.5625661E-02     3.0915655E-02     3.2403336E+00
     0.8000     1.0738885E+01     7.5487345E-01    -2.5722733E-01     6.5897658E-02     3.8301188E-02     4.0706201E+00
     0.9000     1.0446938E+01     1.3971944E+00    -4.5223340E-01     9.2043623E-02     4.7679231E-02     4.4448843E+00
     1.0000     1.0693529E+01     8.3720309E-01    -7.7313137E-01     1.2160468E-01     5.0907865E-02     4.9926033E+00
     1.1000     1.1102855E+01     7.2366279E-01    -6.4175940E-01     1.5406173E-01     5.7760678E-02     5.5162215E+00
     1.2000     1.1778268E+01     2.8858933E-01    -3.7176237E-01     1.9099940E-01     6.4787105E-02     6.1918731E+00
     1.3000     1.1631394E+01     4.3072775E-01    -4.3063450E-01     2.3042324E-01     6.5920517E-02     6.8745251E+00
     1.4000     1.0753760E+01     4.1063738E-01    -8.1212640E-01     2.7107653E-01     7.0706099E-02     6.8937783E+00
     1.5000     1.0745326E+01     7.0404482E-01    -4.8977253E-01     3.1435260E-01     7.1954936E-02     6.8475485E+00
     1.6000     1.0649046E+01    -1.8528491E-01    -4.7523475E-01     3.5746476E-01     7.3262513E-02     6.9930525E+00
     1.7000     1.1410335E+01     7.0972365E-01     3.0976564E-01     4.0341827E-01     7.9529576E-02     7.4091387E+00
     1.8000     1.0567368E+01    -7.4408822E-02    -2.9391944E-01     4.5251366E-01     8.4382720E-02     8.2398758E+00
     1.9000     1.1155082E+01     1.2369564E-01     2.6012611E-01     5.0483292E-01     9.0223797E-02     8.9648695E+00
     2.0000     1.0455896E+01    -2.7935126E-01    -2.7092203E-01     5.6144154E-01     9.8386936E-02     9.4345627E+00
     2.1000     9.8745031E+00    -3.1789616E-01    -6.0941321E-01     6.2195700E-01     1.0296663E-01     1.0010153E+01
     2.2000     1.0448864E+01     3.8147056E-01    -1.2413803E-01     6.8575096E-01     1.1066940E-01     1.0598483E+01
     2.3000     1.0198247E+01     6.0343772E-02    -2.7775595E-01     7.5489980E-01     1.1874805E-01     1.1421090E+01
     2.4000     9.7414551E+00     5.5349410E-01     1.0989508E-01     8.2762766E-01     1.2448428E-01     1.2310097E+01
     2.5000     1.0015521E+01     3.2921234E-01     2.5299061E-03     9.0480876E-01     1.3210678E-01     1.2888032E+01
```

```
*********1*********2*********3*********4*********5*********6*********7*********8
*2345678901234567890123456789012345678901234567890123456789012345678901234567890
********************************************************************************
****************** File generated by Aqwa in Workbench 2024 R1 ****************
********************************************************************************
* Project Title                         :
* Project Reference                     :
* Project Author                        :
* Project Description                   :
* Date of Creation                      : 2025/2/19 16:32:56
* Last Modified                         : 2025/2/19 16:51:36
* This analysis file written at         : 2025/2/19 18:24:44
* Hydrodynamic Solver Unit System : Metric: kg, m [N]
********************************************************************************
*********************************** DECK  0 ************************************
********************************************************************************

DATE:09/03/26      TIME:19:08:27

JOB AQWA  DRIF  WFRQ

TITLE

NUM_CORES         3

OPTIONS CONV FDLL

OPTIONS NOST

OPTIONS REST END

RESTART  1  5


Aqwa-Drift   - 2024 R1                    64BIT VERSION 2024 R1
License Name: aqwa_solve
License Type: Commercial


Point Releases and Patches installed:

Ansys, Inc. License Manager 2024 R1
Viewer 2024 R1
Aqwa 2024 R1
SpaceClaim 2024 R1
Customization Files for User Programmable Features 2024 R1
Mechanical Products 2024 R1
```

4. User guide

Use OpenF2A

Step8: Extract results from the LIS file

Modify the “inputforreadAQWAResults_SingleBody.txt” properly, then double click “readAQWAResults_SingleFile.exe”.

```
! Input file for readAQWAResults_SingleBody which is specific developed for extracting results from the LIS files generated by AQWA.
! The positions, velocities and accelerations of the platform can be output based on the definitions below. The platform is a single
! --------- Global parameters -----------------------
"OC4SEMI_OPENF2A_TURB_114_SW.LIS"                 - InputLISFile       - Input LIS file
"OC4SEMI_OPENF2A_TURB_114_SW"                     - OutFolder      - The target folder to store the result files
3600.0                      - TimeDur
0.1                   - TimeStep
! ---------- Details of the LIS
1                 - NumBodies                                                  le.
!----------- Body 1 ----------------------------------
"Semisub"             - NameBodies(1)      - Name of the 1st body
True                  - FlagOPMot(1)       - Flag of outputting motions at the 1st body's CoG
True                  - FlagOPVel(1)       - Flag of outputting velocities at the 1st body's CoG
True                  - FlagOPAcc(1)       - Flag of outputting accelerations at the 1st body's CoG
True                  - FlagOPFrc(1)       - Flag of outputting forces at the 1st body's CoG
!---------- Added points -----------------------------
0                 - NumAddedPts        - Number of added points that require output resuts (accelerations only)
! --------- Tension of the morring lines ---------------------
True                  - FlagMLType         - Mooring line type (T                                 able )
3                 - NumMoorLines       - Number of mooring lines
! --------- Wave elevation ------------------------
True                  - FlagWvElv      - Output wave elevation
"WAVE PSN NODE 99999-FIXED" - CharWvEle      - Characteristic string of the wave elevation line.
```

Simulation length and time step defined in AQWA

Mooring lines

Wave kinematics at [0,0,0] are extracted by default

Use OpenF2A

Step8: Extract results from the LIS file

Extraction of result will start as follows

```
选择 F:\StudyJeyla\Research\1Papers\2025English\OpenF2A_Case4.0\O
 InputLISFile= OC4SEMI_OPENF2A_TURB_114_SW.LIS
Time =      0 s
Time =    100 s
Time =    200 s
Time =    300 s
Time =    400 s
Time =    500 s
Time =    600 s
Time =    700 s
Time =    800 s
Time =    900 s
Time =   1000 s
Time =   1100 s
Time =   1200 s
Time =   1300 s
Time =   1400 s
Time =   1500 s
Time =   1600 s
Time =   1700 s
Time =   1800 s
Time =   1900 s
Time =   2000 s
Time =   2100 s
Time =   2200 s
Time =   2300 s
```

Data in the LIS file are extracted and written in the files

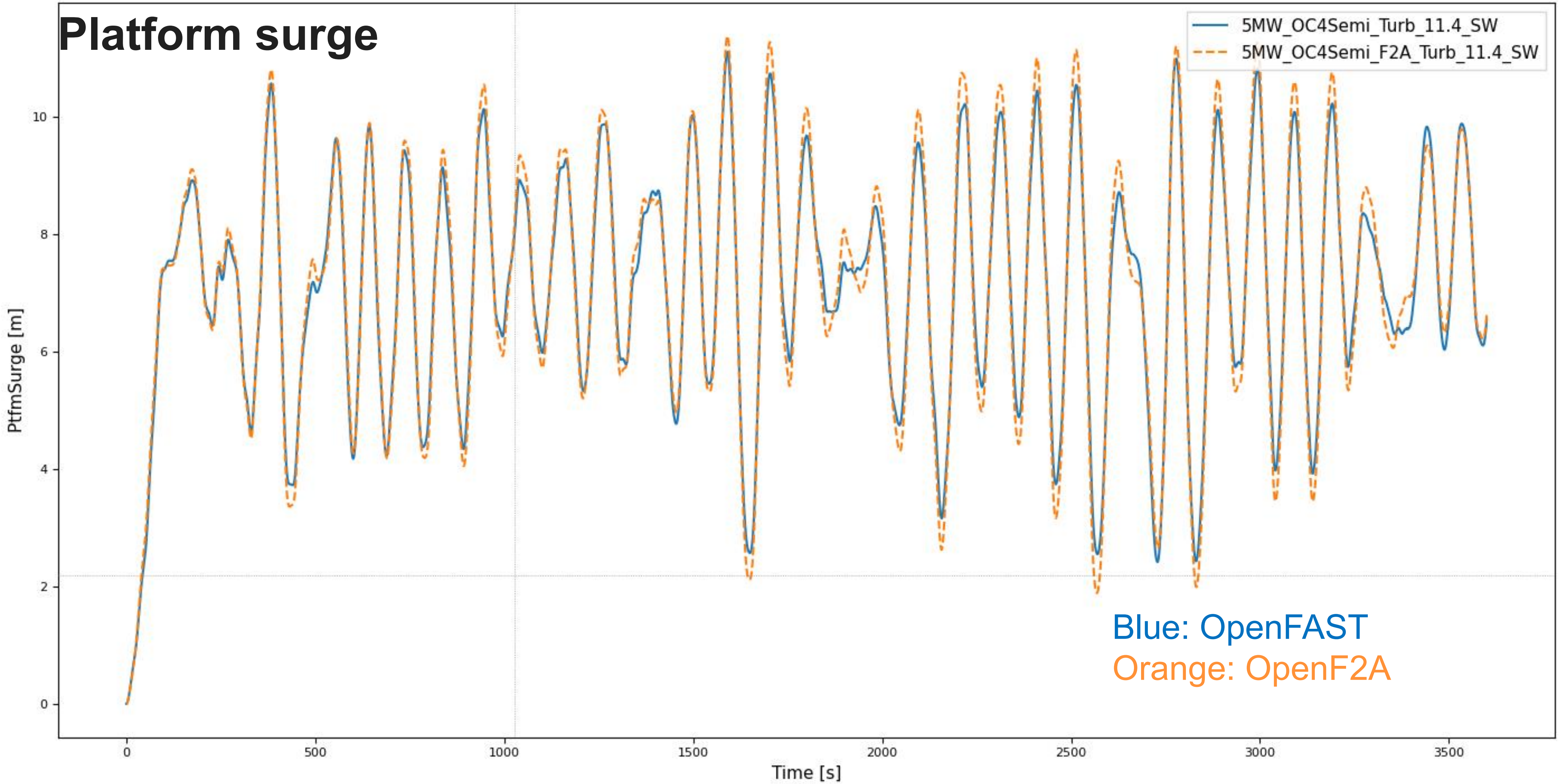
OC4SEMI_OPENF2A_TURB_114_SW

OC4SEMI_OPENF2A_TURB_114_SW_MooringTension.dat	3/9/2026 7:19 PM
OC4SEMI_OPENF2A_TURB_114_SW_Semisub_Acceleration.dat	3/9/2026 7:19 PM
OC4SEMI_OPENF2A_TURB_114_SW_Semisub_Force.dat	3/9/2026 7:19 PM
OC4SEMI_OPENF2A_TURB_114_SW_Semisub_Motion.dat	3/9/2026 7:19 PM
OC4SEMI_OPENF2A_TURB_114_SW_Semisub_Velocity.dat	3/9/2026 7:19 PM
OC4SEMI_OPENF2A_TURB_114_SW_WaveAcceleration.dat	3/9/2026 7:19 PM
OC4SEMI_OPENF2A_TURB_114_SW_WaveElevation.dat	3/9/2026 7:19 PM
OC4SEMI_OPENF2A_TURB_114_SW_WaveVelocity.dat	3/9/2026 7:19 PM

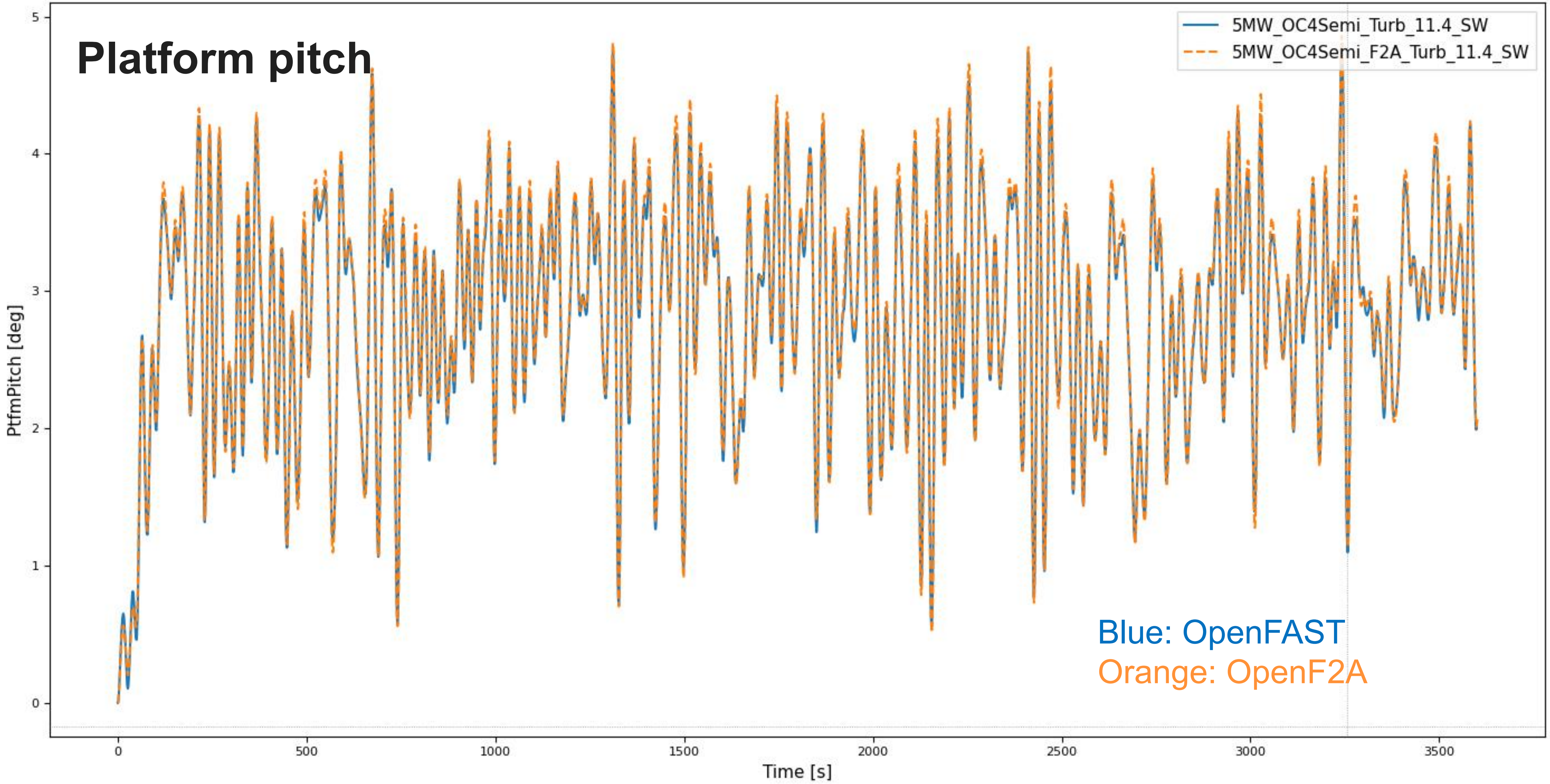
Tension of mooring lines of the OC4SEMI_OPENF2A_TURB_114_SW\OC4SEMI_OPENF2A_TURB_114_SW case extracted

Time s	Mooring1_FL (N)	Mooring2_FL (N)	Mooring3_FL (N)	Mooring1_AC (N)	Mooring2_AC (N)	Mooring3_AC (N)
0.000	1.09640E+06	1.09640E+06	1.09640E+06	8.99110E+05	8.99110E+05	8.99110E+05
0.100	1.09550E+06	1.09570E+06	1.09540E+06	8.99110E+05	8.99110E+05	8.99110E+05
0.200	1.09450E+06	1.09490E+06	1.09430E+06	8.99100E+05	8.99110E+05	8.99100E+05
0.300	1.09350E+06	1.09410E+06	1.09330E+06	8.99020E+05	8.99030E+05	8.99000E+05
0.400	1.09250E+06	1.09350E+06	1.09220E+06	8.98120E+05	8.98280E+05	8.98010E+05
0.500	1.09170E+06	1.09300E+06	1.09130E+06	8.96130E+05	8.96770E+05	8.95890E+05
0.600	1.09090E+06	1.09240E+06	1.09050E+06	8.94440E+05	8.95430E+05	8.94080E+05
0.700	1.08980E+06	1.09170E+06	1.08930E+06	8.92820E+05	8.94360E+05	8.92320E+05
0.800	1.08790E+06	1.09010E+06	1.08730E+06	8.91420E+05	8.93610E+05	8.90840E+05
0.900	1.08610E+06	1.08880E+06	1.08540E+06	8.90350E+05	8.92830E+05	8.89750E+05
1.000	1.08480E+06	1.08790E+06	1.08390E+06	8.89330E+05	8.92220E+05	8.88680E+05
1.100	1.08370E+06	1.08750E+06	1.08280E+06	8.87980E+05	8.91100E+05	8.87230E+05
1.200	1.08300E+06	1.08750E+06	1.08210E+06	8.86520E+05	8.89930E+05	8.85690E+05
1.300	1.08260E+06	1.08750E+06	1.08160E+06	8.85670E+05	8.89510E+05	8.84770E+05
1.400	1.08200E+06	1.08750E+06	1.08090E+06	8.85320E+05	8.89520E+05	8.84330E+05
1.500	1.08120E+06	1.08710E+06	1.08010E+06	8.85220E+05	8.90180E+05	8.84230E+05
1.600	1.08100E+06	1.08710E+06	1.07980E+06	8.85450E+05	8.90930E+05	8.84420E+05
1.700	1.08150E+06	1.08770E+06	1.08020E+06	8.85590E+05	8.91670E+05	8.84520E+05
1.800	1.08230E+06	1.08850E+06	1.08100E+06	8.85540E+05	8.92180E+05	8.84380E+05

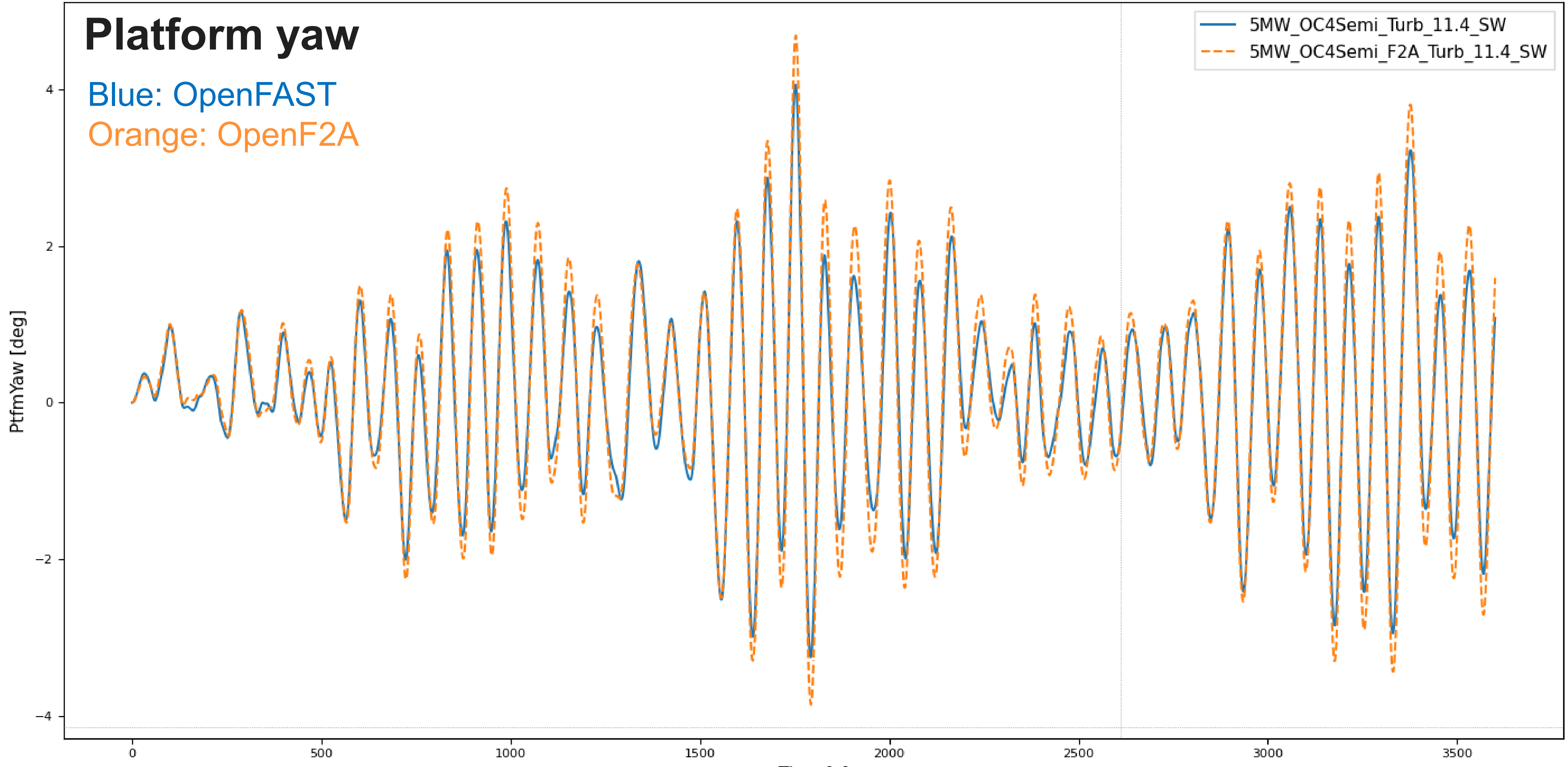
Comparison between OpenFAST and OpenF2A



Comparison between OpenFAST and OpenF2A



Comparison between OpenFAST and OpenF2A



References

The following publications are for your information when you need a reference to cite if OpenF2A is useful in your research. The 1st and 8th publications listed below are recommended the most. We are grateful for your acknowledgement by citing our research. More suitable references can be found on my google scholar website https://scholar.google.com/citations?user=frTAwNQAAAAJ&hl=zh-CN.

1.Yang, Y., Yu, J., Bashir, M., & Li, C. (2026). Development and applicability of a novel fully coupled framework for floating wind energy systems. Ocean Engineering, 354, 124900

2.Hu, G., Ding, J., Lai, Y., He, C., Zhang, Z., Liu, W., ... & Yang, Y. (2026). Coupled dynamic behavior of a floating offshore wind turbine integrated with flapping-type-wave energy converters under wind-wave misalignment conditions. Ocean Engineering, 353, 124816.

3.Liu, W., ZHANG, Z., Lai, Y., Hu, G., Yu, J., Li, C., ... & Yang, Y. (2026). Stability and power performance of a floating wind-current energy system under tidal turbine operating mode variations. Energy, 140209.

4.Nie, D., Li, S., Yan, Y., Lou, Y., Yin, J., Hao, J., ... & Yang, Y. (2025). Structural analysis of a 15 MW floating offshore wind turbine platform based on a novel fully-coupled framework. Ocean Engineering, 342, 123057.

5.Yin, J., Fan, Y., Bashir, M., Nie, D., Lai, Y., Ding, J., ... & Yang, Y. (2025). Development of a hybrid deep learning model with HHO algorithm for dynamic response prediction of wind-wave integrated floating energy systems. Ocean Engineering, 340, 122394.

6.Ding, J., Yang, Y., Yu, J., Bashir, M., Ma, L., Li, C., & Li, S. (2024). Fully coupled dynamic responses of barge-type integrated floating wind-wave energy systems with different WEC layouts. Ocean Engineering, 313, 119453.

7.Yang, Y., Shi, Z., Fu, J., Ma, L., Yu, J., Fang, F., ... & Yang, W. (2023). Effects of tidal turbine number on the performance of a 10 MW-class semi-submersible integrated floating wind-current system. Energy, 285, 128789.

8.Yang, Y., Bashir, M., Michailides, C., Li, C., & Wang, J. (2020). Development and application of an aero-hydro-servo-elastic coupling framework for analysis of floating offshore wind turbines. Renewable Energy, 161, 606-625.

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Thank you!